Assessment on LAB-04

Design and simulate a **3-bit synchronous UP/DOWN counter** using Verilog in ModelSim. The counter should have:

- **clk** – clock input
- **reset** – active-high synchronous reset
- **enable** – counter enable input
- **up_down** – counting direction control input
- **Q** – 3-bit counter output

reset	enable	up_down	Operation
1	X	X	Reset counter to 000
0	0	X	Hold the current count
0	1	1	Count upward
0	1	0	Count downward

EE 4122

Exp. : 05

Implementation of different logical operations using VHDL for designing FPGA device using Quartus II software.

Boolean expression: $Y = (A + BC)'$

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity circuit1 is
    Port (
        A : in  STD_LOGIC;
        B : in  STD_LOGIC;
        C : in  STD_LOGIC;
        Y : out STD_LOGIC
    );
end circuit1;

architecture Behavioral of circuit1 is
begin

    Y <= not (A or (B and C));

end Behavioral;
```

$Y = A \oplus B \oplus C$

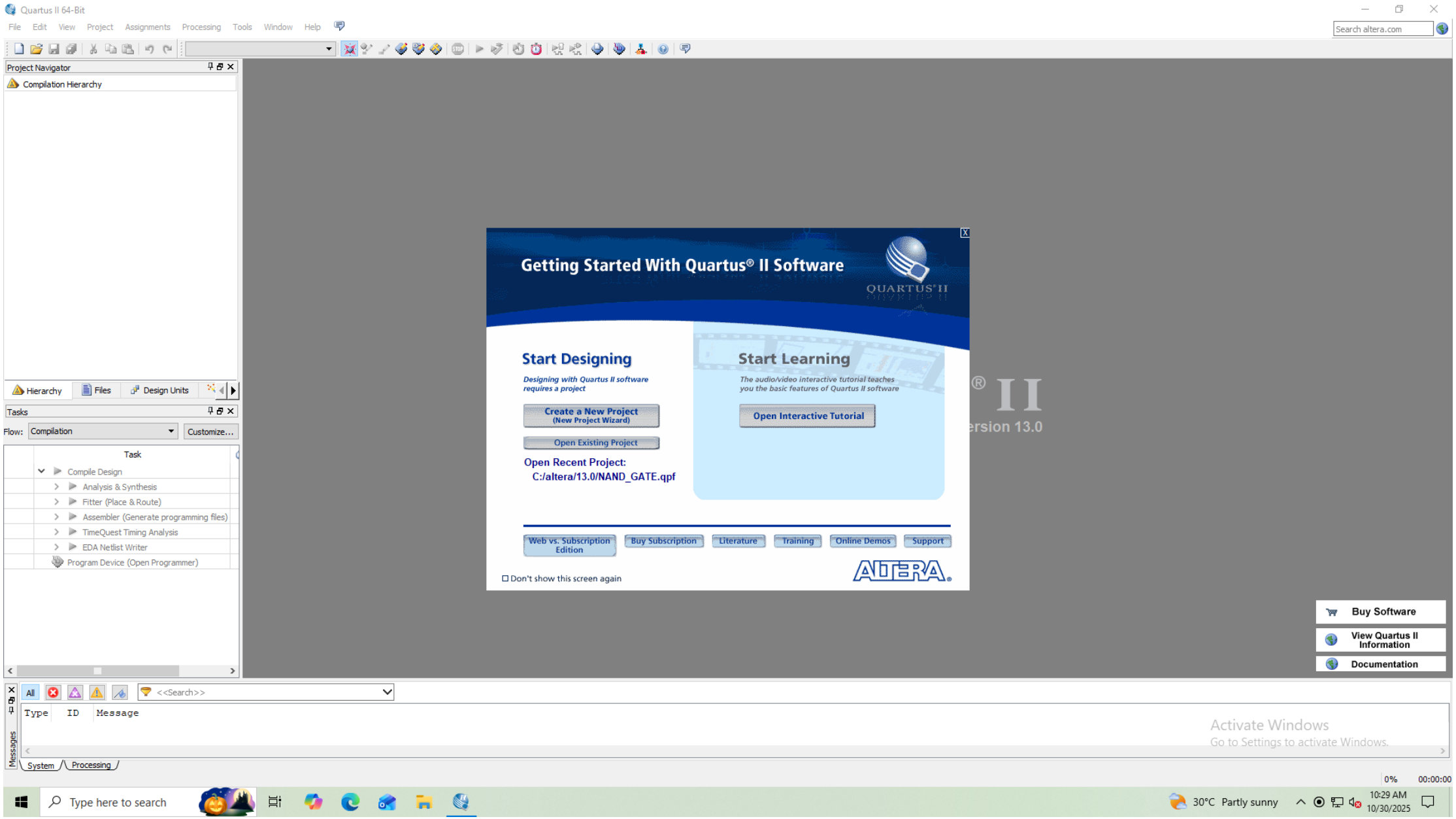
```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity circuit2 is
    Port (
        A : in  STD_LOGIC;
        B : in  STD_LOGIC;
        C : in  STD_LOGIC;
        Y : out STD_LOGIC
    );
end circuit2;

architecture Behavioral of circuit2 is
begin

    Y <= A xor B xor C;

end Behavioral;
```


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File Edit View Project Assignments Processing Tools Window Help
Search altera.com
Project Navigator
Compilation Hierarchy
Hierarchy Files Design Units
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Flow: Compilation Customize...
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TimeQuest Timing Analysis
EDA Netlist Writer
Program Device (Open Programmer)
Getting Started With Quartus® II Software
QUARTUS II
Start Designing
Designing with Quartus II software requires a project
Create a New Project (New Project Wizard)
Open Existing Project
Open Recent Project:
C:/altera/13.0/NAND_GATE.qpf
Start Learning
The audio/video interactive tutorial teaches you the basic features of Quartus II software
Open Interactive Tutorial
Web vs. Subscription Edition
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10/30/2025

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New... Ctrl+N
Open... Ctrl+O
Close Ctrl+F4

New Project Wizard...
Open Project... Ctrl+J
Save Project
Close Project

Save Ctrl+S
Save As...
Save All Ctrl+Shift+S

File Properties...

Create / Update
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Convert Programming Files...

Page Setup...
Print Preview
Print... Ctrl+P

Recent Files
Recent Projects

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Flow: Compilation Customize...

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Messages

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Project Navigator

Compilation Hierarchy

Hierarchy Files Design Units

Tasks

Flow: Compilation Customize...

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EDA Netlist Writer
Program Device (Open Programmer)

New Project Wizard

Introduction

The New Project Wizard helps you create a new project and preliminary project settings, including the following:

- Project name and directory
- Name of the top-level design entity
- Project files and libraries
- Target device family and device
- EDA tool settings

You can change the settings for an existing project and specify additional project-wide settings with the Settings command (Assignments menu). You can use the various pages of the Settings dialog box to add functionality to the project.

Don't show me this introduction again

< Back Next > Finish Cancel Help

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Type	ID	Message

Messages

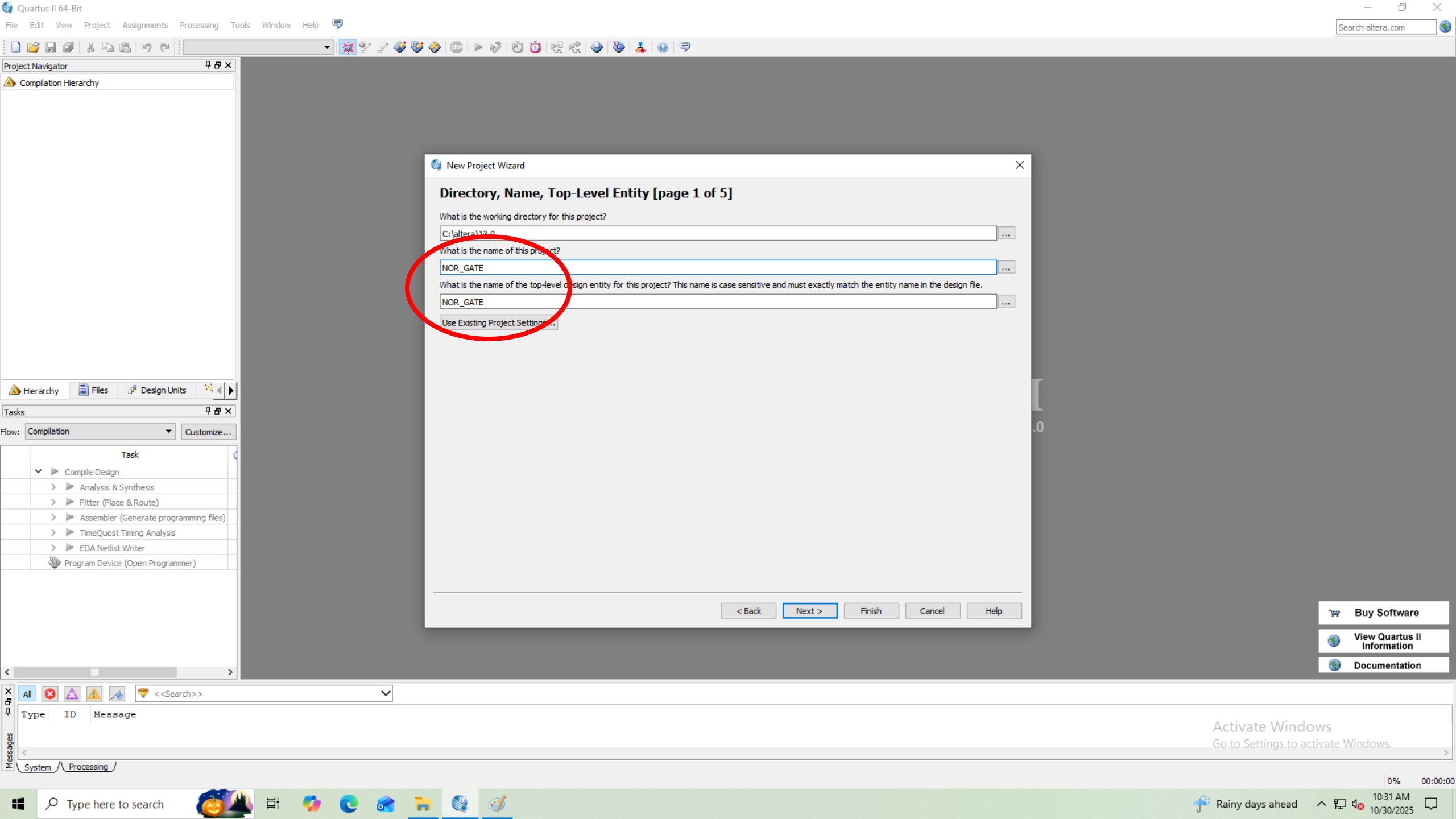
System Processing

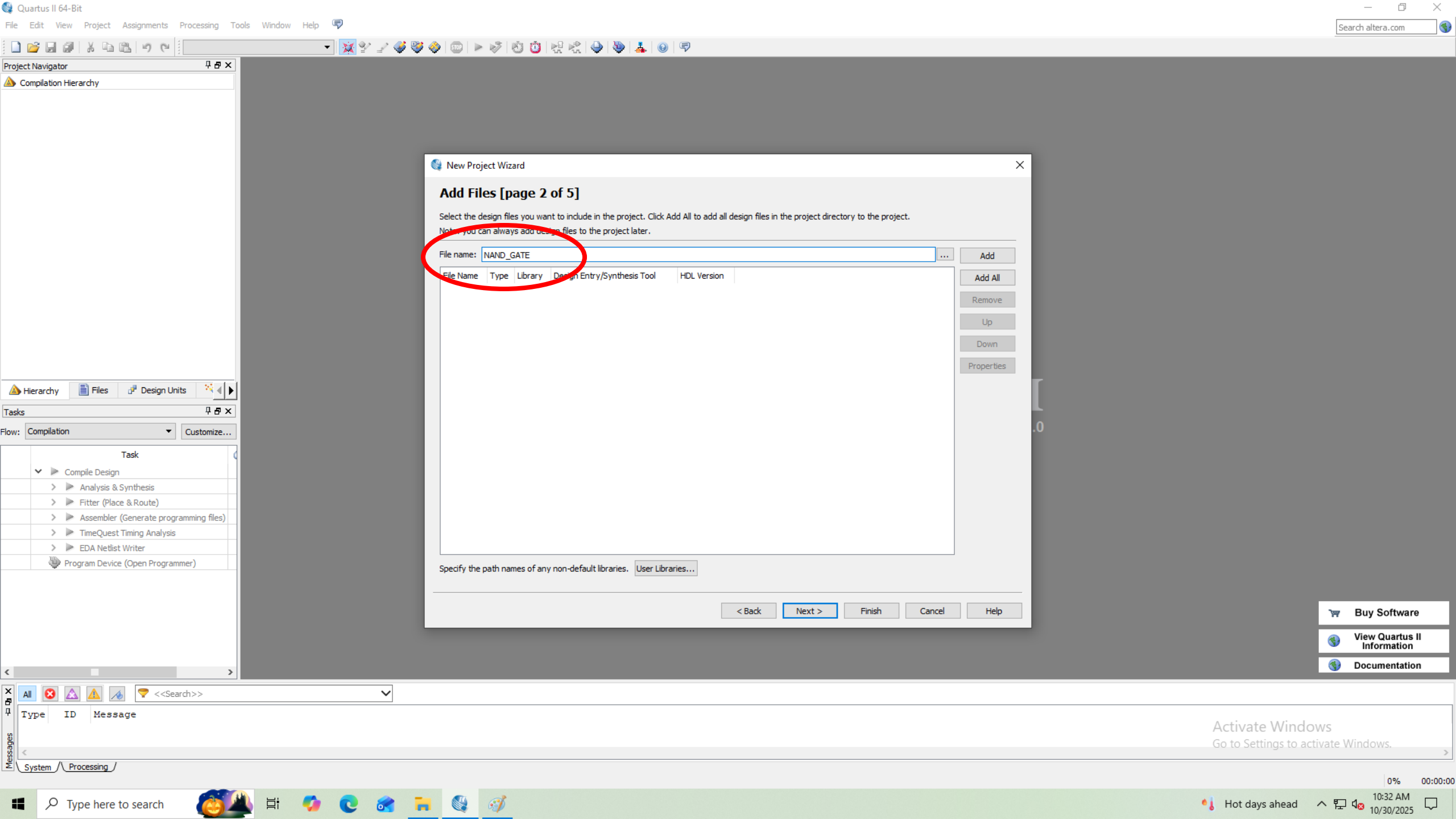
Activate Windows
Go to Settings to activate Windows.

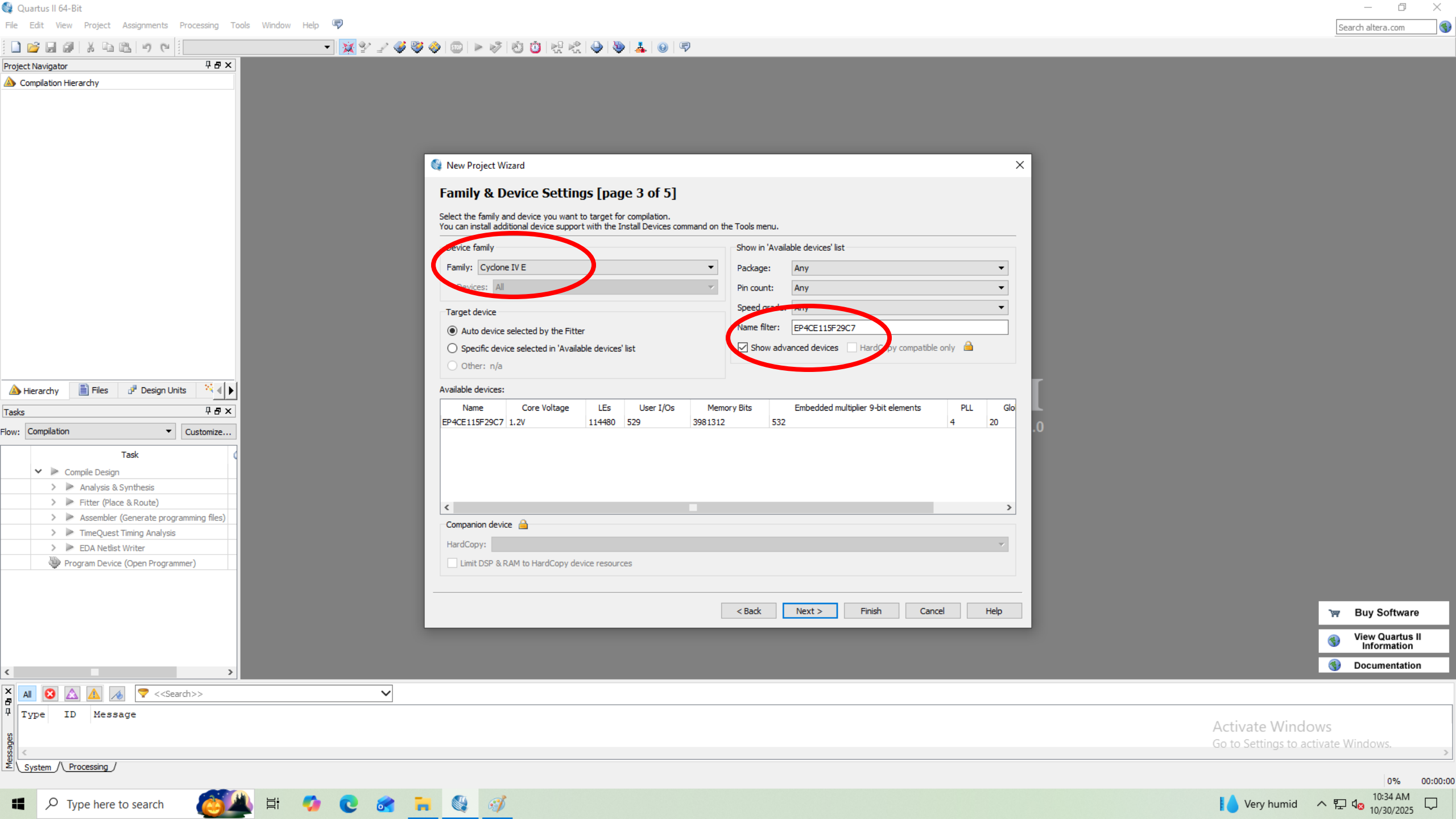
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File Edit View Project Assignments Processing Tools Window Help
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Compilation Hierarchy
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New Project Wizard
Directory, Name, Top-Level Entity [page 1 of 5]
What is the working directory for this project?
C:\altera\13.0
What is the name of this project?
NOR_GATE
What is the name of the top-level design entity for this project? This name is case sensitive and must exactly match the entity name in the design file.
NOR_GATE
Use Existing Project Settings...
< Back
Next >
Finish
Cancel
Help
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New Project Wizard
Add Files [page 2 of 5]
Select the design files you want to include in the project. Click Add All to add all design files in the project directory to the project.
Note: you can always add design files to the project later.
File name: NAND_GATE
Add
Add All
Remove
Up
Down
Properties
File Name Type Library Design Entry/Synthesis Tool HDL Version
Specify the path names of any non-default libraries. User Libraries...
< Back Next > Finish Cancel Help
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File Edit View Project Assignments Processing Tools Window Help
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Project Navigator
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Program Device (Open Programmer)
New Project Wizard
Family & Device Settings [page 3 of 5]
Select the family and device you want to target for compilation.
You can install additional device support with the Install Devices command on the Tools menu.
Device family
Family: Cyclone IV E
Devices: All
Target device
Auto device selected by the Fitter
Specific device selected in 'Available devices' list
Other: n/a
Show in 'Available devices' list
Package: Any
Pin count: Any
Speed grade: Any
Name filter: EP4CE115F29C7
Show advanced devices
HardCopy compatible only
Available devices:
Name Core Voltage LEs User I/Os Memory Bits Embedded multiplier 9-bit elements PLL Glo
EP4CE115F29C7 1.2V 114480 529 3981312 532 4 20
Companion device
HardCopy:
Limit DSP & RAM to HardCopy device resources
< Back Next > Finish Cancel Help
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Very humid
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10/30/2025

Quartus II 64-Bit - C:/altera/13.0/OR_GATE - OR_GATE

File Edit View Project Assignments Processing Tools Window Help

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New... Ctrl+N
Open... Ctrl+O
Close Ctrl+F4
New Project Wizard...
Open Project... Ctrl+J
Save Project
Close Project
Save Ctrl+S
Save As...
Save All Ctrl+Shift+S
File Properties...
Create / Update
Export...
Convert Programming Files...
Page Setup...
Print Preview
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Recent Files
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Exit Alt+F4

IP Components

Flow: Compilation Customize...

Task	Time
Compile Design	
Analysis & Synthesis	
Fitter (Place & Route)	
Assembler (Generate programming files)	
TimeQuest Timing Analysis	
EDA Netlist Writer	
Program Device (Open Programmer)	

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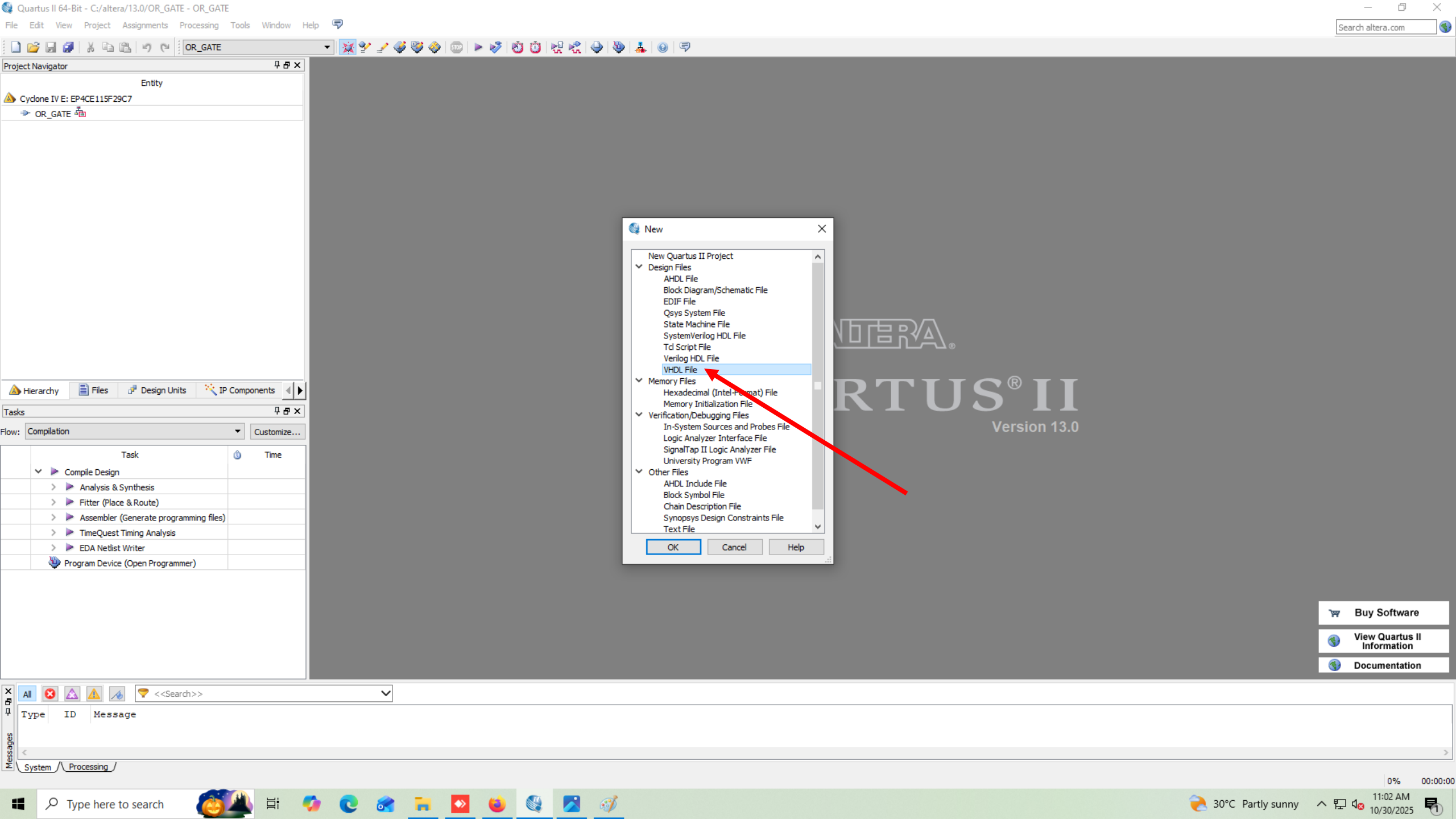
Messages System Processing

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Quartus II 64-Bit - C:/altera/13.0/OR_GATE - OR_GATE
File Edit View Project Assignments Processing Tools Window Help
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Cyclone IV E: EP4CE115F29C7
OR_GATE
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Tasks
Flow: Compilation
Customize...
Task Time
Compile Design
Analysis & Synthesis
Fitter (Place & Route)
Assembler (Generate programming files)
TimeQuest Timing Analysis
EDA Netlist Writer
Program Device (Open Programmer)
New
New Quartus II Project
Design Files
AHDL File
Block Diagram/Schematic File
EDIF File
Qsys System File
State Machine File
SystemVerilog HDL File
Tcl Script File
Verilog HDL File
VHDL File
Memory Files
Hexadecimal (Intel-Format) File
Memory Initialization File
Verification/Debugging Files
In-System Sources and Probes File
Logic Analyzer Interface File
SignalTap II Logic Analyzer File
University Program VWF
Other Files
AHDL Include File
Block Symbol File
Chain Description File
Synopsys Design Constraints File
Text File
OK Cancel Help
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Quartus II 64-Bit - C:/altera/13.0/OR_GATE - OR_GATE

File Edit View Project Assignments Processing Tools Window Help

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OR_GATE

Project Navigator

Entity

Cyclone IV E: EP4CE115F29C7

OR_GATE

Hierarchy Files Design Units IP Components

Tasks

Flow: Compilation Customize...

Task	Time
Compile Design	
Analysis & Synthesis	
Fitter (Place & Route)	
Assembler (Generate programming files)	
TimeQuest Timing Analysis	
EDA Netlist Writer	
Program Device (Open Programmer)	

Vhdl1.vhd*

```
1  library IEEE;
2  use IEEE.STD_LOGIC_1164.ALL;
3
4  entity OR_GATE is
5      Port (
6          A : in  STD_LOGIC;
7          B : in  STD_LOGIC;
8          Y : out STD_LOGIC
9      );
10 end OR_GATE;
11
12 architecture Behavioral of OR_GATE is
13 begin
14     Y <= A OR B;
15 end Behavioral;
```

All <<Search>>

Type ID Message

Messages

System Processing

Ln 15 Col 16 VHDL File 0% 00:00:00

Type here to search

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Quartus II 64-Bit - C:/altera/13.0/OR_GATE - OR_GATE

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Open... Ctrl+O
Close Ctrl+F4
New Project Wizard...
Open Project... Ctrl+J
Save Project
Close Project
Save Ctrl+S
Save As...
Save All Ctrl+Shift+S
File Properties...
Create / Update
Export...
Convert Programming Files...
Page Setup...
Print Preview
Print... Ctrl+P
Recent Files
Recent Projects
Exit Alt+F4

OR_GATE.vhd

```
1  library IEEE;
2  use IEEE.STD_LOGIC_1164.ALL;
3
4  entity OR_GATE is
5      Port (
6          A : in  STD_LOGIC;
7          B : in  STD_LOGIC;
8          Y : out STD_LOGIC
9      );
10 end OR_GATE;
11
12 architecture Behavioral of OR_GATE is
13 begin
14     Y <= A OR B;
15 end Behavioral;
```

IP Components

Flow: Compilation Customize...

Task	Time
Compile Design	
Analysis & Synthesis	
Fitter (Place & Route)	
Assembler (Generate programming files)	
TimeQuest Timing Analysis	
EDA Netlist Writer	
Program Device (Open Programmer)	

All <<Search>>

Type ID Message

Messages System Processing

Saves all open files and the current project's settings and assignments

0% 00:00:00

Type here to search

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11:05 AM 10/30/2025

Quartus II 64-Bit - C:/altera/13.0/OR_GATE - OR_GATE

File Edit View Project Assignments Processing Tools Window Help

Search altera.com

OR_GATE

Project Navigator

Entity

Cyclone IV E: EP4CE115F29C7

OR_GATE

Hierarchy Files Design Units IP Components

Tasks

Flow: Compilation Customize...

	Task	Time
	Compile Design	
✓	Analysis & Synthesis	00:00:02
	Fitter (Place & Route)	
	Assembler (Generate programming files)	
	TimeQuest Timing Analysis	
	EDA Netlist Writer	
	Program Device (Open Programmer)	

OR_GATE.vhd

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- Flow Non-Default Global Settings
- Flow Elapsed Time
- Flow OS Summary
- Flow Log
- Analysis & Synthesis
- Flow Messages
- Flow Suppressed Messages

Compilation Report - OR_GATE

Flow Summary

Flow Status	Successful - Thu Oct 30 11:05:35 2025
Quartus II 64-Bit Version	13.0.0 Build 156 04/24/2013 SJ Web Edition
Revision Name	OR_GATE
Top-level Entity Name	OR_GATE
Family	Cyclone IV E
Device	EP4CE115F29C7
Timing Models	Final
Total logic elements	1
Total combinational functions	1
Dedicated logic registers	0
Total registers	0
Total pins	3
Total virtual pins	0
Total memory bits	0
Embedded Multiplier 9-bit elements	0
Total PLLs	0

All <<Search>>

Type	ID	Message
		Quartus II 64-Bit Analysis & Synthesis was successful. 0 errors, 1 warning

Messages

System Processing (10)

100% 00:00:02

Type here to search

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11:05 AM 10/30/2025

Quartus II 64-Bit - C:/altera/13.0/OR_GATE - OR_GATE

File Edit View Project Assignments Processing Tools Window Help

Search altera.com

OR_GATE

Project Navigator

Entity

Cyclone IV E: EP4CE115F29C7

OR_GATE

Hierarchy | Files | Design Units | IP Components

Tasks

Flow: Compilation | Customize...

	Task	Time
✓	Compile Design	00:00:16
✓	Analysis & Synthesis	00:00:02
✓	Fitter (Place & Route)	00:00:08
✓	Assembler (Generate programming files)	00:00:03
✓	TimeQuest Timing Analysis	00:00:02
✓	EDA Netlist Writer	00:00:01
	Program Device (Open Programmer)	

OR_GATE.vhd

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- Assembler
- TimeQuest Timing Analyzer
- EDA Netlist Writer
- Flow Messages
- Flow Suppressed Messages

Compilation Report - OR_GATE

Flow Summary

Flow Status	Successful - Thu Oct 30 11:06:22 2025
Quartus II 64-Bit Version	13.0.0 Build 156 04/24/2013 SJ Web Edition
Revision Name	OR_GATE
Top-level Entity Name	OR_GATE
Family	Cyclone IV E
Device	EP4CE115F29C7
Timing Models	Final
Total logic elements	1 / 114,480 (< 1 %)
Total combinational functions	1 / 114,480 (< 1 %)
Dedicated logic registers	0 / 114,480 (0 %)
Total registers	0
Total pins	3 / 529 (< 1 %)
Total virtual pins	0
Total memory bits	0 / 3,981,312 (0 %)
Embedded Multiplier 9-bit elements	0 / 532 (0 %)
Total PLLs	0 / 4 (0 %)

All <<Search>>

Type	ID	Message
	293000	Quartus II Full Compilation was successful. 0 errors, 12 warnings

Messages

System | Processing (125)

100% 00:00:16

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11:06 AM 10/30/2025

Quartus II 64-Bit - C:/altera/13.0/OR_GATE - OR_GATE

File Edit View Project Assignments Processing Tools Window Help

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OR_GATE

Project Navigator

Entity

Cyclone IV E: EP4CE115F29C7

OR_GATE

Hierarchy Files Design Units IP Components

Tasks

Flow: Compilation Customize...

	Task	Time
✓	Compile Design	00:00:16
✓	Analysis & Synthesis	00:00:02
✓	Fitter (Place & Route)	00:00:08
✓	Assembler (Generate programming files)	00:00:03
✓	TimeQuest Timing Analysis	00:00:02
✓	EDA Netlist Writer	00:00:01
	Program Device (Open Programmer)	

OR_GATE.vhd

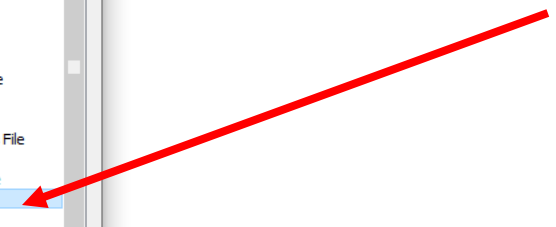
```
1  library IEEE;
2  use IEEE.STD_LOGIC_1164.ALL;
3
4  entity OR_GATE is
5      Port (
6          A : in  STD_LOGIC;
7          B : in  STD_LOGIC;
8          Y : out STD_LOGIC
9      );
10 end OR_GATE;
11
12 architecture Behavioral of OR_GATE is
13 begin
14     Y <= A OR B;
15 end Behavioral;
```

Compilation Report - OR_GATE

New

- New Quartus II Project
- Design Files
  - AHDL File
  - Block Diagram/Schematic File
  - EDIF File
  - Qsys System File
  - State Machine File
  - SystemVerilog HDL File
  - Tcl Script File
  - Verilog HDL File
  - VHDL File
- Memory Files
  - Hexadecimal (Intel-Format) File
  - Memory Initialization File
- Verification/Debugging Files
  - In-System Sources and Probes File
  - Logic Analyzer Interface File
  - SignalTap II Logic Analyzer File
  - University Program VWF
- Other Files
  - AHDL Include File
  - Block Symbol File
  - Chain Description File
  - Synopsys Design Constraints File
  - Text File

OK Cancel Help



All <<Search>>

Type	ID	Message
	293000	Quartus II Full Compilation was successful. 0 errors, 12 warnings

Messages

System Processing (125)

100% 00:00:16

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Simulation Waveform Editor - C:/altera/13.0/OR_GATE - OR_GATE - [Waveform1.vwf]

File Edit View Simulation Help

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Open... Ctrl+O
Close Ctrl+F4
Save Ctrl+S
Save As...

Pointer: 4.74 ns Interval: 4.74 ns Start: End:

0 ps 40.0 ns 80.0 ns 120.0 ns 160.0 ns 200.0 ns 240.0 ns 280.0 ns 320.0 ns 360.0 ns 400.0 ns 440.0 ns 480.0 ns 520.0 ns 560.0 ns 600.0 ns 640.0 ns 680.0 ns 720.0 ns 760.0 ns 800.0 ns 840.0 ns 880.0 ns 920.0 ns 960.0 ns 1.0 us

0 ps

Save the waveform with new name

0% 00:00:00

Type here to search

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Simulation Waveform Editor - C:/altera/13.0/OR_GATE - OR_GATE - [Waveform_OR.vwf]

File Edit View Simulation Help

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Delete Del

Insert

Value

Grouping

Reverse Group or Bus Bit Order

Radix

Grid Size...

Set End Time...

Snap to Grid

Snap to Transition

Properties...

Insert Node or Bus...

Pointer: 4.15 ns Interval: 4.15 ns Start: End:

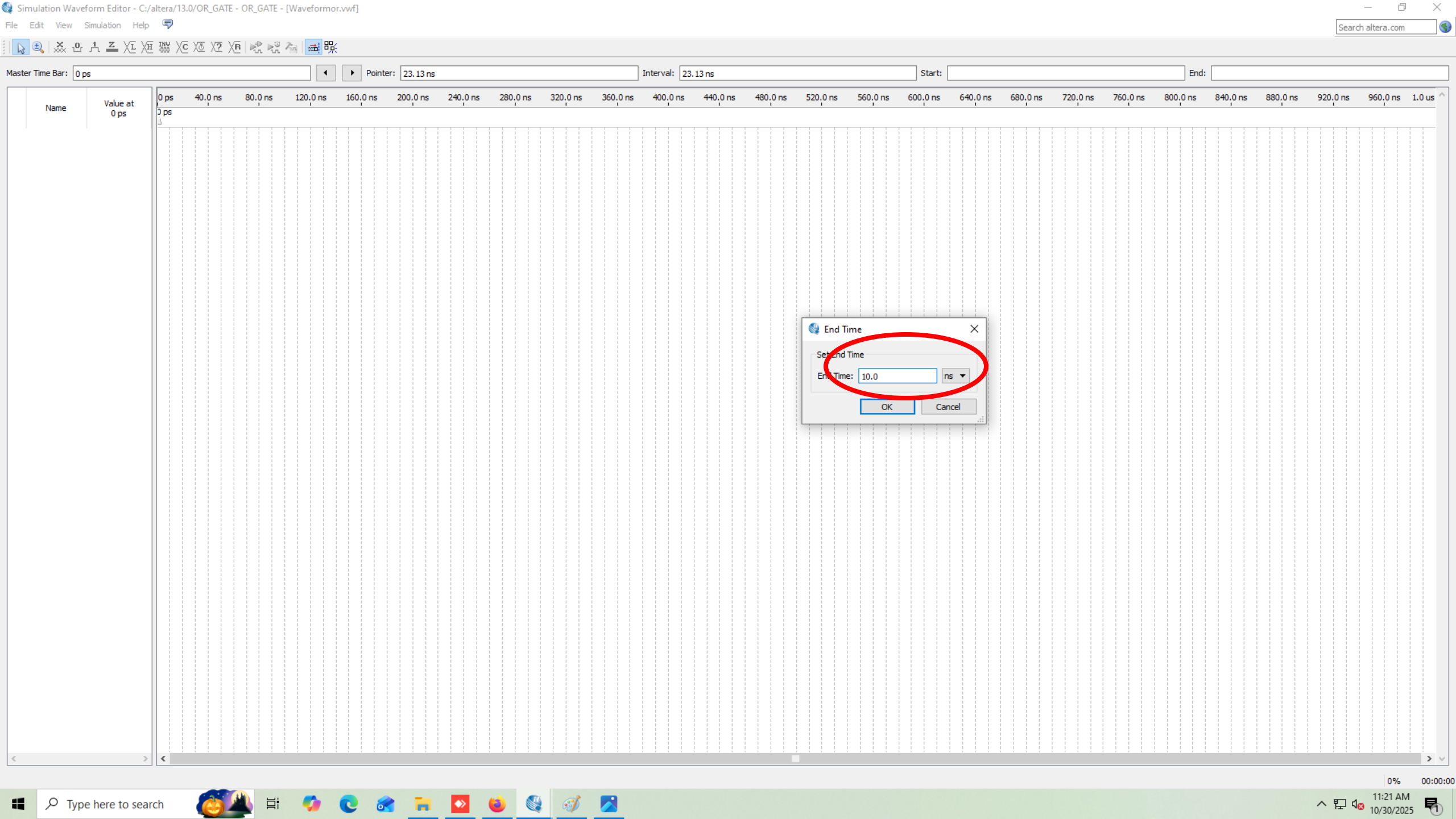
0.0 ns 80.0 ns 120.0 ns 160.0 ns 200.0 ns 240.0 ns 280.0 ns 320.0 ns 360.0 ns 400.0 ns 440.0 ns 480.0 ns 520.0 ns 560.0 ns 600.0 ns 640.0 ns 680.0 ns 720.0 ns 760.0 ns 800.0 ns 840.0 ns 880.0 ns 920.0 ns 960.0 ns 1.0 us

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Simulation Waveform Editor - C:/altera/13.0/OR_GATE - OR_GATE - [Waveformor.vwf]
File Edit View Simulation Help
Search altera.com
Master Time Bar: 0 ps
Pointer: 23.13 ns
Interval: 23.13 ns
Start:
End:
Name
Value at 0 ps
0 ps 40.0 ns 80.0 ns 120.0 ns 160.0 ns 200.0 ns 240.0 ns 280.0 ns 320.0 ns 360.0 ns 400.0 ns 440.0 ns 480.0 ns 520.0 ns 560.0 ns 600.0 ns 640.0 ns 680.0 ns 720.0 ns 760.0 ns 800.0 ns 840.0 ns 880.0 ns 920.0 ns 960.0 ns 1.0 us
End Time
Set End Time
End Time: 10.0 ns
OK
Cancel
0%
00:00:00
Type here to search
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Simulation Waveform Editor - C:/altera/13.0/OR_GATE - OR_GATE - [Waveformor.vwf]*

File Edit View Simulation Help

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Delete Del

Insert

Value

Grouping

Reverse Group or Bus Bit Order

Radix

Grid Size...

Set End Time...

Snap to Grid

Snap to Transition

Properties...

Pointer: 225 ps Interval: 225 ps Start: End:

Change the File's grid settings

0% 00:00:00

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Simulation Waveform Editor - C:/altera/13.0/OR_GATE - OR_GATE - [Waveform_OR.vwf]

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Insert

Value

Grouping

Reverse Group or Bus Bit Order

Radix

Grid Size...

Set End Time...

Snap to Grid

Snap to Transition

Properties...

Insert Node or Bus...

Pointer: 4.15 ns Interval: 4.15 ns Start: End:

80.0 ns 120.0 ns 160.0 ns 200.0 ns 240.0 ns 280.0 ns 320.0 ns 360.0 ns 400.0 ns 440.0 ns 480.0 ns 520.0 ns 560.0 ns 600.0 ns 640.0 ns 680.0 ns 720.0 ns 760.0 ns 800.0 ns 840.0 ns 880.0 ns 920.0 ns 960.0 ns 1.0 us

0% 00:00:00

Type here to search

Rainy days ahead

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Simulation Waveform Editor - C:/altera/13.0/OR_GATE - OR_GATE - [Waveform_OR.vwf]

File Edit View Simulation Help

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Master Time Bar: 0 ps Pointer: 12.45 ns Interval: 12.45 ns Start: End:

Name | Value at 0 ps

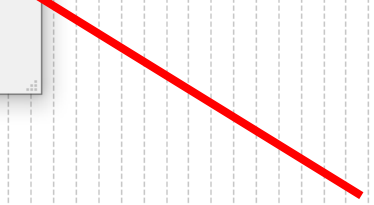
0 ps 40.0 ns 80.0 ns 120.0 ns 160.0 ns 200.0 ns 240.0 ns 280.0 ns 320.0 ns 360.0 ns 400.0 ns 440.0 ns 480.0 ns 520.0 ns 560.0 ns 600.0 ns 640.0 ns 680.0 ns 720.0 ns 760.0 ns 800.0 ns 840.0 ns 880.0 ns 920.0 ns 960.0 ns 1.0 us

0 ps

Insert Node or Bus

Name:
Type: INPUT
Value type: 9-Level
Radix: Binary
Bus width: 1
Start index: 0
Display gray code count as binary count

OK
Cancel
Node Finder...

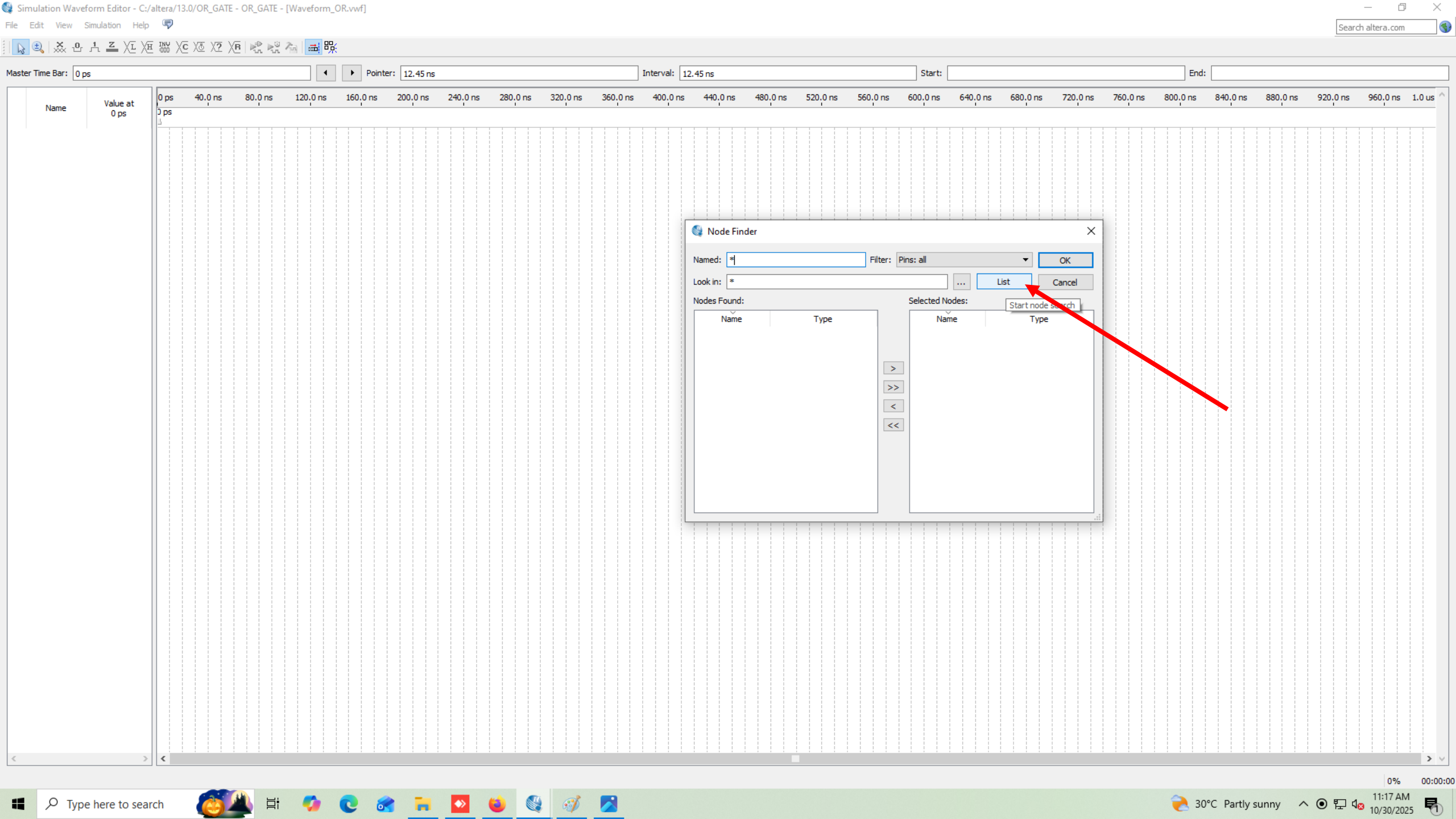


0% 00:00:00

Type here to search

Rainy days ahead

11:10 AM 10/30/2025


Simulation Waveform Editor - C:/altera/13.0/OR_GATE - OR_GATE - [Waveform_OR.vwf]
File Edit View Simulation Help
Search altera.com
Master Time Bar: 0 ps
Pointer: 12.45 ns
Interval: 12.45 ns
Start:
End:
Name
Value at 0 ps
0 ps 40.0 ns 80.0 ns 120.0 ns 160.0 ns 200.0 ns 240.0 ns 280.0 ns 320.0 ns 360.0 ns 400.0 ns 440.0 ns 480.0 ns 520.0 ns 560.0 ns 600.0 ns 640.0 ns 680.0 ns 720.0 ns 760.0 ns 800.0 ns 840.0 ns 880.0 ns 920.0 ns 960.0 ns 1.0 us
Node Finder
Named: *
Filter: Pins: all
OK
Look in: *
...
List
Cancel
Start node search
Nodes Found:
Name
Type
Selected Nodes:
Name
Type
>
>>
<
<<
0%
00:00:00
Type here to search
30°C Partly sunny
11:17 AM
10/30/2025

Simulation Waveform Editor - C:/altera/13.0/OR_GATE - OR_GATE - [Waveform_OR.vwf]

File Edit View Simulation Help

Search altera.com

Master Time Bar: 0 ps Pointer: 12.45 ns Interval: 12.45 ns Start: End:

Name | Value at 0 ps

0 ps 40.0 ns 80.0 ns 120.0 ns 160.0 ns 200.0 ns 240.0 ns 280.0 ns 320.0 ns 360.0 ns 400.0 ns 440.0 ns 480.0 ns 520.0 ns 560.0 ns 600.0 ns 640.0 ns 680.0 ns 720.0 ns 760.0 ns 800.0 ns 840.0 ns 880.0 ns 920.0 ns 960.0 ns 1.0 us

0 ps

Node Finder

Named: * Filter: Pins: all OK

Look in: * ... List Cancel

Nodes Found:

Name	Type
A	Input
B	Input
Y	Output

Selected Nodes:

Name	Type
A	Input
B	Input
Y	Output

> >> < <<

Copy all to Selected Nodes list

0% 00:00:00

Type here to search

30°C Partly sunny 11:17 AM 10/30/2025

Simulation Waveform Editor - C:/altera/13.0/OR_GATE - OR_GATE - [Waveform_OR.vwf]*

File Edit View Simulation Help

Search altera.com

Master Time Bar: 0 ps Pointer: 53.37 ns Interval: 53.37 ns Start: 0 ps End: 1.0 us

Name	Value at 0 ps
A	B 0
B	B 0
Y	B X

0 ps 40.0 ns 80.0 ns 120.0 ns 160.0 ns 200.0 ns 240.0 ns 280.0 ns 320.0 ns 360.0 ns 400.0 ns 440.0 ns 480.0 ns 520.0 ns 560.0 ns 600.0 ns 640.0 ns 680.0 ns 720.0 ns 760.0 ns 800.0 ns 840.0 ns 880.0 ns 920.0 ns 960.0 ns 1.0 us

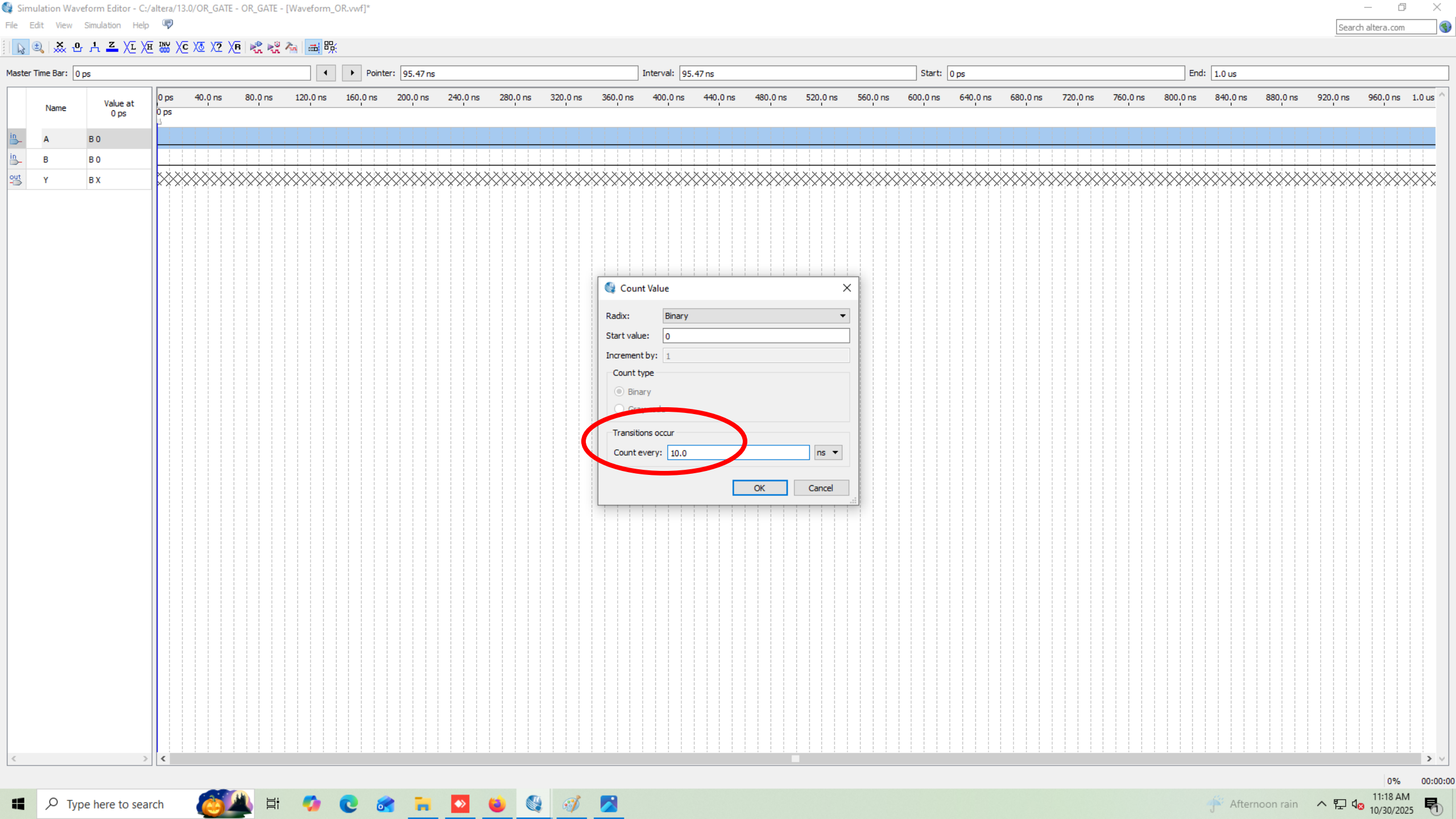
Overwrite Count Value

0% 00:00:00

Type here to search

Afternoon rain

11:17 AM 10/30/2025


Simulation Waveform Editor - C:/altera/13.0/OR_GATE - OR_GATE - [Waveform_OR.vwf]*
File Edit View Simulation Help
Search altera.com
Master Time Bar: 0 ps
Pointer: 95.47 ns
Interval: 95.47 ns
Start: 0 ps
End: 1.0 us
Name
Value at 0 ps
A
B 0
B
B 0
Y
B X
0 ps 40.0 ns 80.0 ns 120.0 ns 160.0 ns 200.0 ns 240.0 ns 280.0 ns 320.0 ns 360.0 ns 400.0 ns 440.0 ns 480.0 ns 520.0 ns 560.0 ns 600.0 ns 640.0 ns 680.0 ns 720.0 ns 760.0 ns 800.0 ns 840.0 ns 880.0 ns 920.0 ns 960.0 ns 1.0 us
Count Value
Radix: Binary
Start value: 0
Increment by: 1
Count type
Binary
Transitions occur
Count every: 10.0
ns
OK
Cancel
0%
00:00:00
Type here to search
Afternoon rain
11:18 AM
10/30/2025

Simulation Waveform Editor - C:/altera/13.0/OR_GATE - OR_GATE - [Waveform_OR.vwf]*

File Edit View Simulation Help

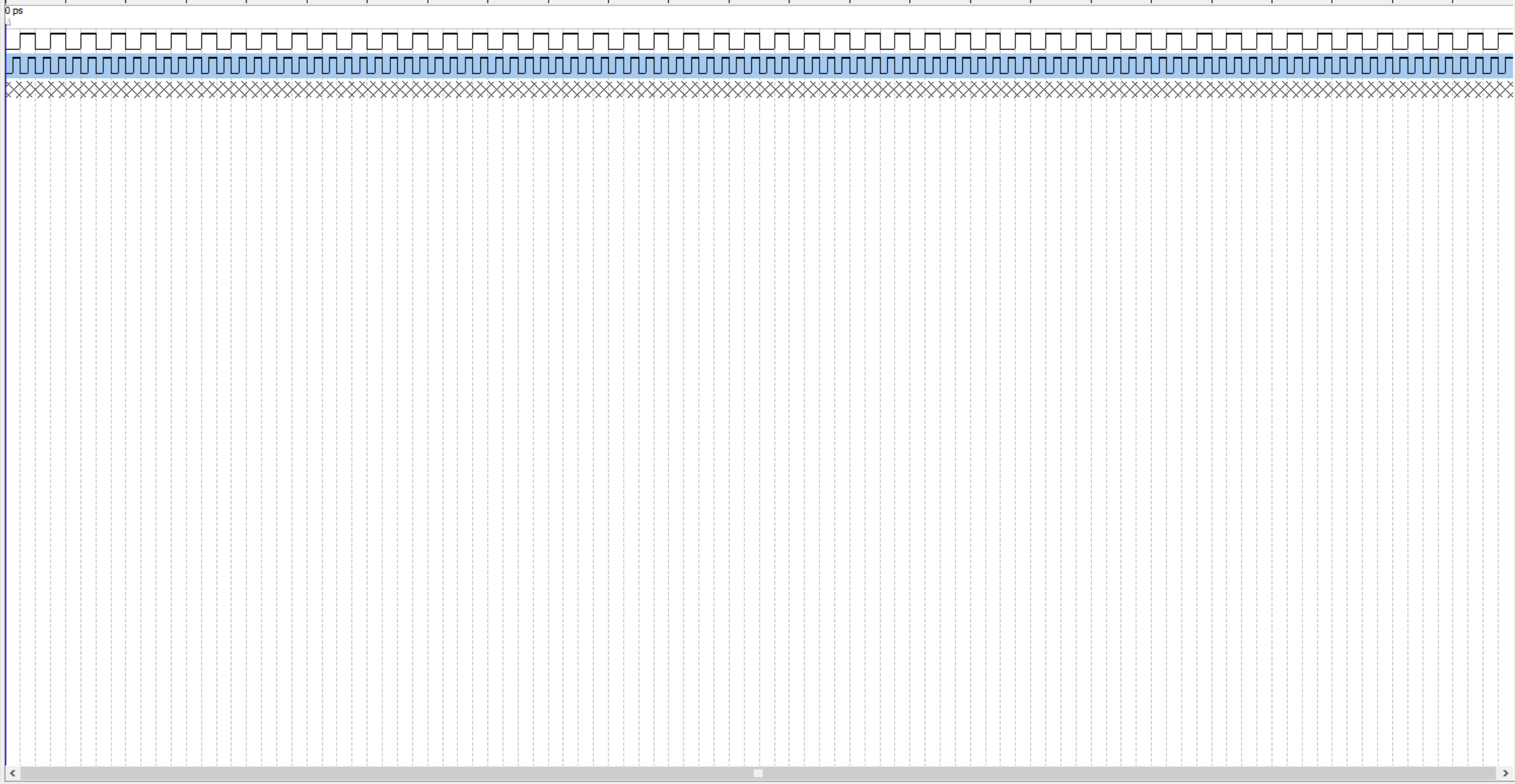
Search altera.com

Run Functional Simulation

Master Time Bar: 0 ps Pointer: 71.16 ns Interval: 71.16 ns Start: 0 ps End: 1.0 us

Name	Value at 0 ps
A	B 0
B	B 0
Y	B X

0 ps 40.0 ns 80.0 ns 120.0 ns 160.0 ns 200.0 ns 240.0 ns 280.0 ns 320.0 ns 360.0 ns 400.0 ns 440.0 ns 480.0 ns 520.0 ns 560.0 ns 600.0 ns 640.0 ns 680.0 ns 720.0 ns 760.0 ns 800.0 ns 840.0 ns 880.0 ns 920.0 ns 960.0 ns 1.0 us



Run a functional simulation 0% 00:00:00

Type here to search Afternoon rain 11:19 AM 10/30/2025

Simulation Waveform Editor - C:/altera/13.0/OR_GATE - OR_GATE - [Waveformor.vwf]*

File Edit View Simulation Help

Search altera.com

Open... Ctrl+O
Close Ctrl+F4
Save Ctrl+S
Save As...

Pointer: 0 ps Interval: 0 ps Start: 0 ps End: 10.0 ns

in	A	B 0
in	B	B 0
out	Y	B X

0 ps 500 ps 1.0 ns 1.5 ns 2.0 ns 2.5 ns 3.0 ns 3.5 ns 4.0 ns 4.5 ns 5.0 ns 5.5 ns 6.0 ns 6.5 ns 7.0 ns 7.5 ns 8.0 ns

Save the waveform

0% 00:00:00

Type here to search

Rain coming

11:24 AM
10/30/2025

Simulation Waveform Editor - C:/altera/13.0/AND_GATE - AND_GATE - [Waveform1.vwf]

File Edit View Simulation Help

Search altera.com

Options

Run Functional Simulation

Run Timing Simulation

Generate ModelSim Testbench and Script

Master Time Bar: Pointer: 405 ps Interval: 405 ps Start: 0 ps End: 10.0 ns

Name	
A	B 0
B	B 0
Y	B X

0 ps 1.0 ns 1.5 ns 2.0 ns 2.5 ns 3.0 ns 3.5 ns 4.0 ns 4.5 ns 5.0 ns 5.5 ns 6.0 ns 6.5 ns 7.0 ns 7.5 ns 8.0 ns

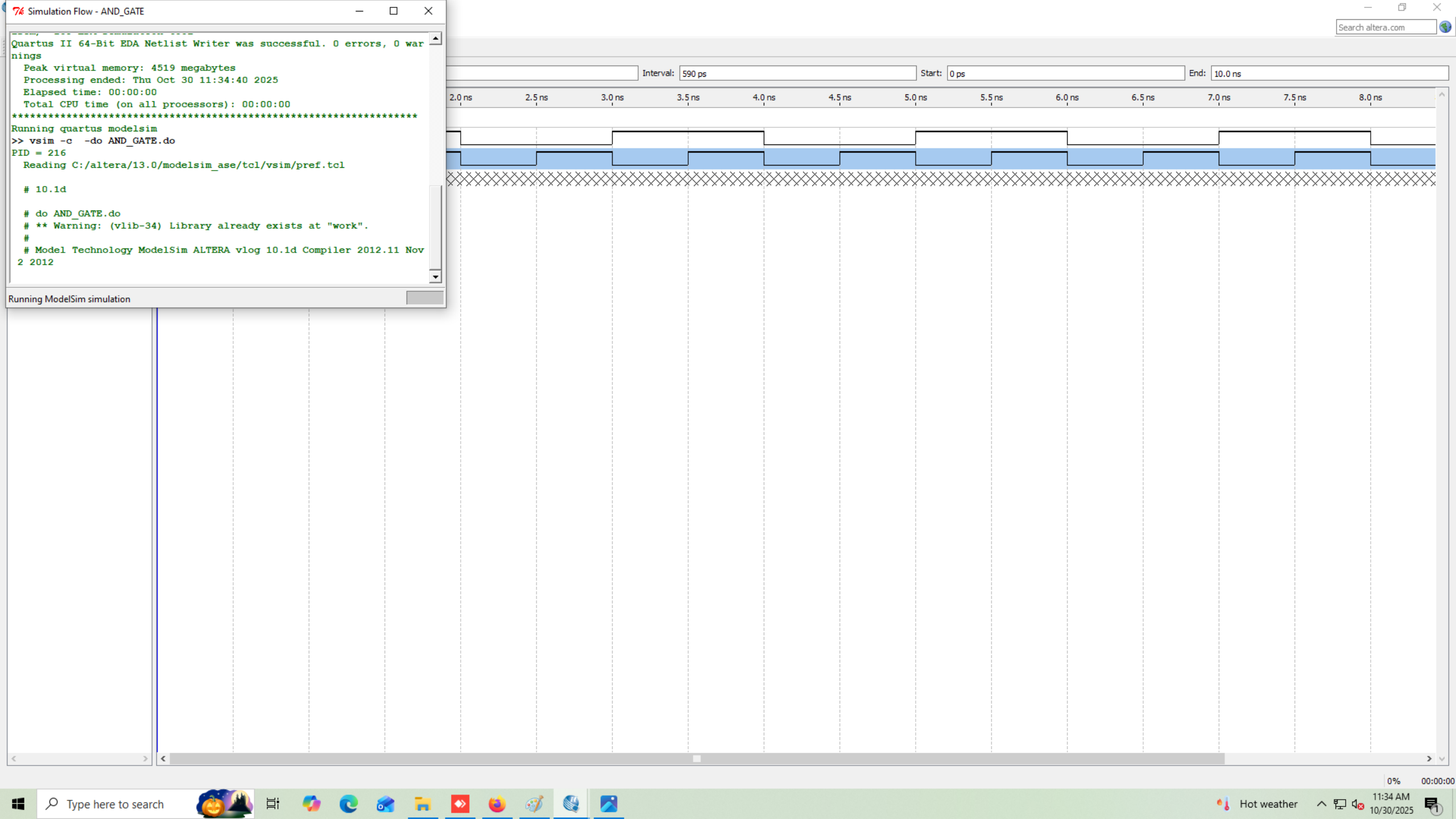
Run a functional simulation

0% 00:00:00

Type here to search

Hot weather

11:35 AM
10/30/2025


Simulation Flow - AND_GATE
Quartus II 64-Bit EDA Netlist Writer was successful. 0 errors, 0 warnings
Peak virtual memory: 4519 megabytes
Processing ended: Thu Oct 30 11:34:40 2025
Elapsed time: 00:00:00
Total CPU time (on all processors): 00:00:00
Running quartus modelsim
>> vsim -c -do AND_GATE.do
PID = 216
Reading C:/altera/13.0/modelsim_ase/tcl/vsim/pref.tcl
10.1d
do AND_GATE.do
** Warning: (vlib-34) Library already exists at "work".
Model Technology ModelSim ALTERA vlog 10.1d Compiler 2012.11 Nov 2 2012
Running ModelSim simulation
Interval: 590 ps
Start: 0 ps
End: 10.0 ns
2.0 ns
2.5 ns
3.0 ns
3.5 ns
4.0 ns
4.5 ns
5.0 ns
5.5 ns
6.0 ns
6.5 ns
7.0 ns
7.5 ns
8.0 ns
Search altera.com
Type here to search
Hot weather
11:34 AM
10/30/2025

Simulation Waveform Editor - C:/altera/13.0/AND_GATE - AND_GATE - [Waveform1.vwf]

File Edit View Simulation Help

Search altera.com

Master Time Bar: 0 ps | Pointer: 5.08 ns | Interval: 5.08 ns | Start: 0 ps | End: 10.0 ns

Name	Value at 0 ps
A	B 0
B	B 0
Y	B X

0 ps 500 ps 1.0 ns 1.5 ns 2.0 ns 2.5 ns 3.0 ns 3.5 ns 4.0 ns 4.5 ns 5.0 ns 5.5 ns 6.0 ns 6.5 ns 7.0 ns 7.5 ns 8.0 ns

0 ps

Simulation Waveform Editor - [AND_GATE.sim.vwf (Read-Only)]

File Edit View Simulation Help

Search altera.com

Master Time Bar: 0 ps | Pointer: | Interval: | Start: | End:

Name	Value at 0 ps
A	B 0
B	B 0
Y	B 0

0 ps

0 ps

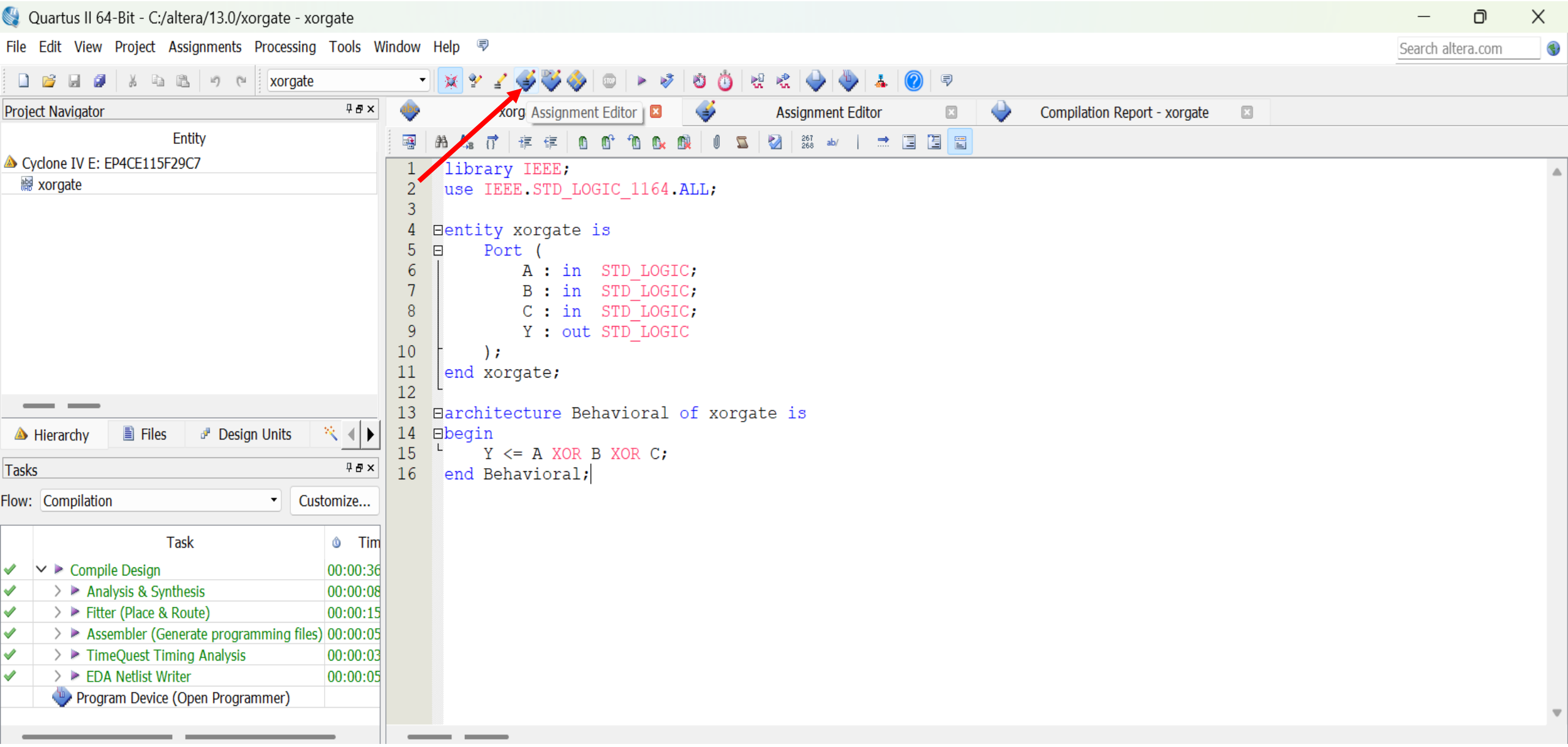
0% 00:00:00

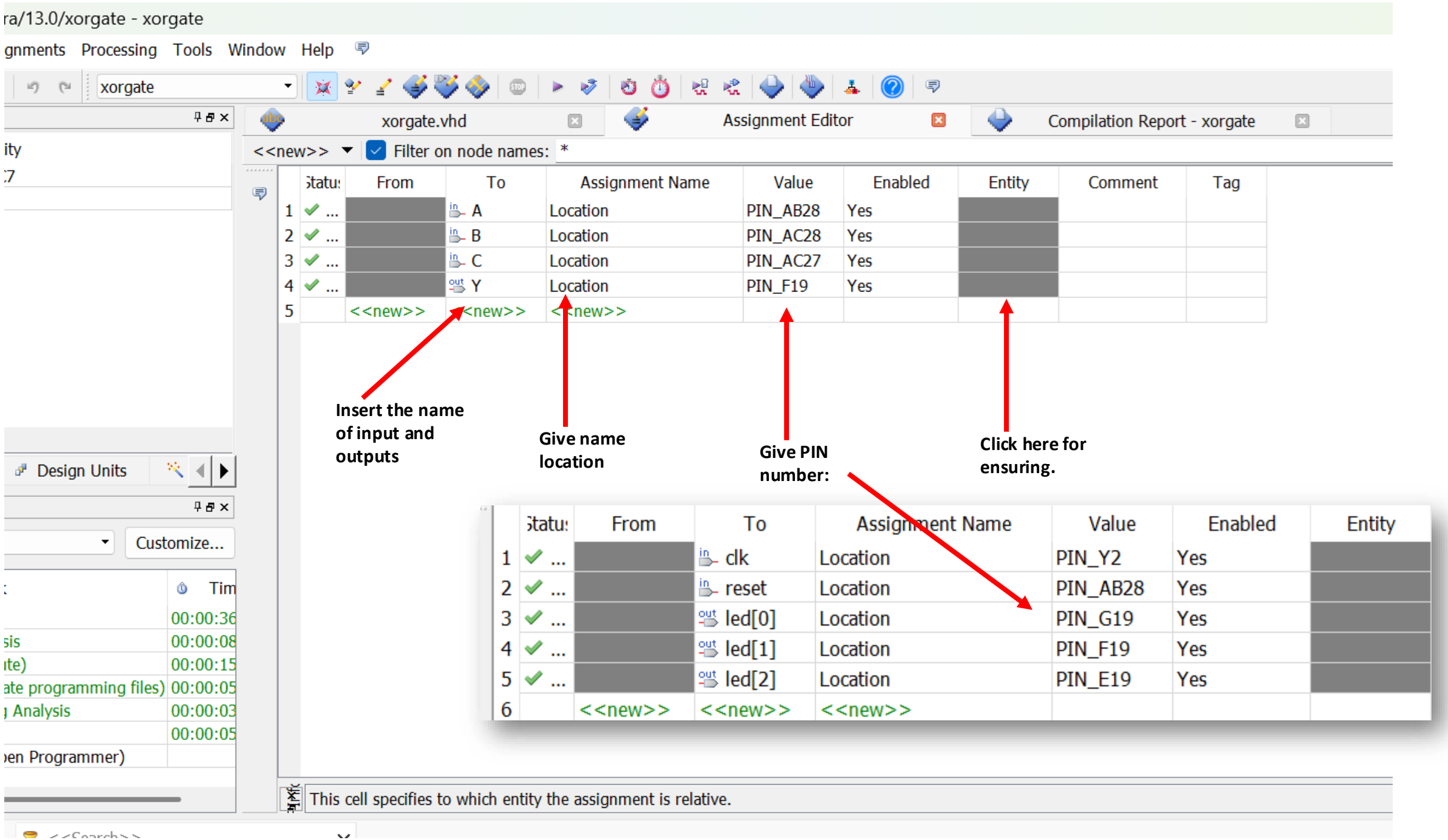
0% 00:00:00

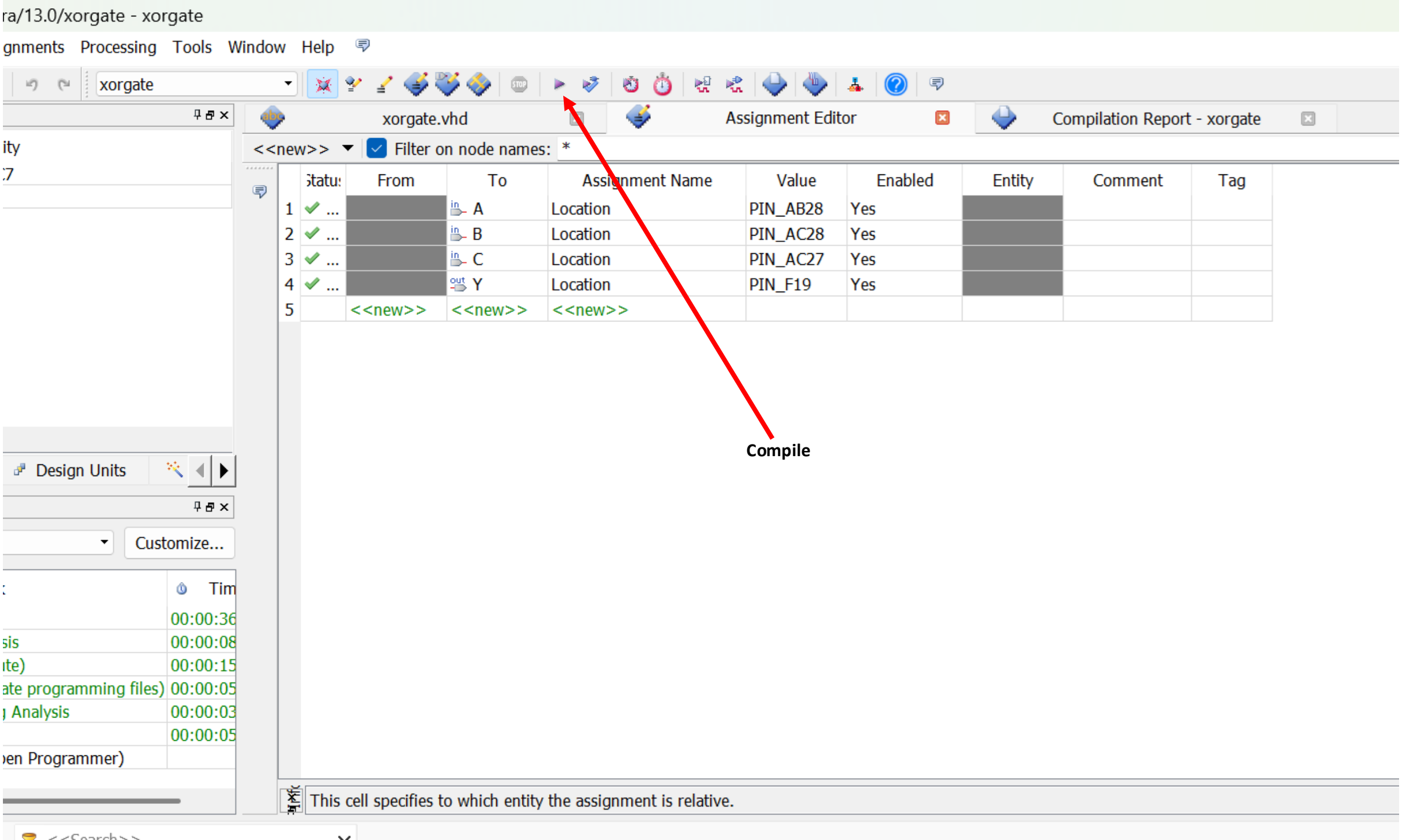
Type here to search

Hot weather

11:34 AM
10/30/2025


Quartus II 64-Bit - C:/altera/13.0/xorgate - xorgate
File Edit View Project Assignments Processing Tools Window Help
Search altera.com
xorgate
Project Navigator
Entity
Cyclone IV E: EP4CE115F29C7
xorgate
Assignment Editor
Assignment Editor
Compilation Report - xorgate
1 library IEEE;
2 use IEEE.STD_LOGIC_1164.ALL;
3
4 entity xorgate is
5 Port (
6 A : in STD_LOGIC;
7 B : in STD_LOGIC;
8 C : in STD_LOGIC;
9 Y : out STD_LOGIC
10);
11 end xorgate;
12
13 architecture Behavioral of xorgate is
14 begin
15 Y <= A XOR B XOR C;
16 end Behavioral;
Hierarchy
Files
Design Units
Tasks
Flow: Compilation
Customize...
Task
Tim
Compile Design 00:00:36
Analysis & Synthesis 00:00:08
Fitter (Place & Route) 00:00:15
Assembler (Generate programming files) 00:00:05
TimeQuest Timing Analysis 00:00:03
EDA Netlist Writer 00:00:05
Program Device (Open Programmer)


ra/13.0/xorgate - xorgate
gnments Processing Tools Window Help
xorgate
xorgate.vhd
Assignment Editor
Compilation Report - xorgate
<<new>> Filter on node names: *
Status From To Assignment Name Value Enabled Entity Comment Tag
1 ... A Location PIN_AB28 Yes
2 ... B Location PIN_AC28 Yes
3 ... C Location PIN_AC27 Yes
4 ... Y Location PIN_F19 Yes
5 <<new>> <<new>> <<new>>
Insert the name of input and outputs
Give name location
Give PIN number:
Click here for ensuring.
Status From To Assignment Name Value Enabled Entity
1 ... clk Location PIN_Y2 Yes
2 ... reset Location PIN_AB28 Yes
3 ... led[0] Location PIN_G19 Yes
4 ... led[1] Location PIN_F19 Yes
5 ... led[2] Location PIN_E19 Yes
6 <<new>> <<new>> <<new>>
Design Units
Customize...
This cell specifies to which entity the assignment is relative.


ra/13.0/xorgate - xorgate
gnments Processing Tools Window Help
xorgate
xorgate.vhd
Assignment Editor
Compilation Report - xorgate
<<new>>
Filter on node names: *
Status From To Assignment Name Value Enabled Entity Comment Tag
1 ... A Location PIN_AB28 Yes
2 ... B Location PIN_AC28 Yes
3 ... C Location PIN_AC27 Yes
4 ... Y Location PIN_F19 Yes
5 <<new>> <<new>> <<new>>
Compile
Design Units
Customize...
Tim
00:00:36
00:00:08
00:00:15
00:00:05
00:00:03
00:00:05
This cell specifies to which entity the assignment is relative.

Quartus II 64-Bit - C:/altera/13.0/xorgate - xorgate

File Edit View Project Assignments Processing Tools Window Help

xorgate

Project Navigator

Entity

Cyclone IV E: EP4CE115F29C7

xorgate

Hierarchy Files Design Units

Tasks

Flow: Compilation Customize...

	Task	Tim
26%	Compile Design	00:00:04
✓	Analysis & Synthesis	00:00:02
30%	Fitter (Place & Route)	00:00:02
0%	Assembler (Generate programming files)	00:00:00
0%	TimeQuest Timing Analysis	00:00:00
0%	EDA Netlist Writer	00:00:00
	Program Device (Open Programmer)	

xorgate.vhd | Assignment Editor | Compilation Report - xorgate

Table ...

Flow Sur
Flow Set
Flow Nor
Flow Ela
Flow OS
Flow Log
Analysis
Fitter
Flow Mes
Flow Sup

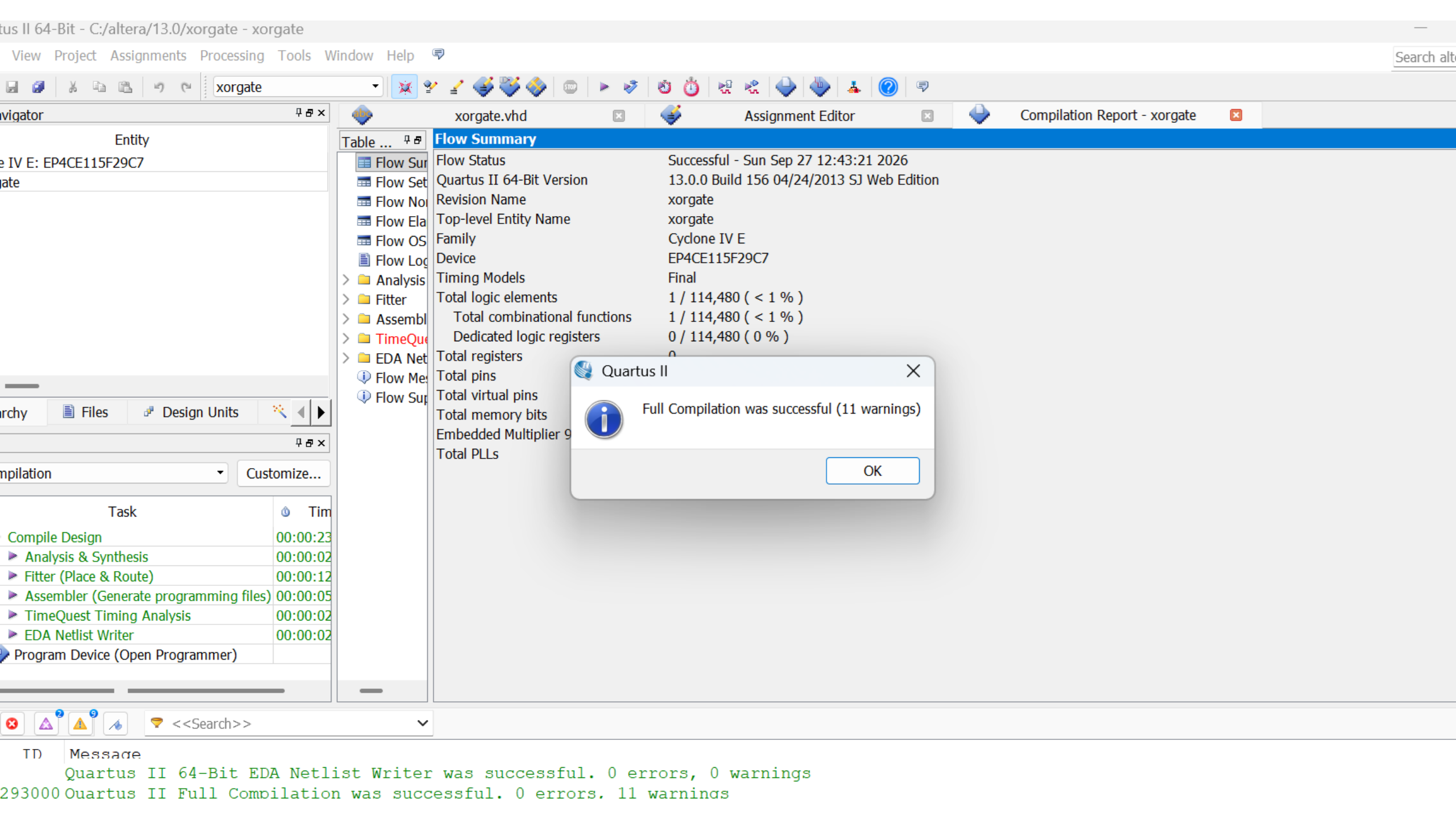
Flow Summary

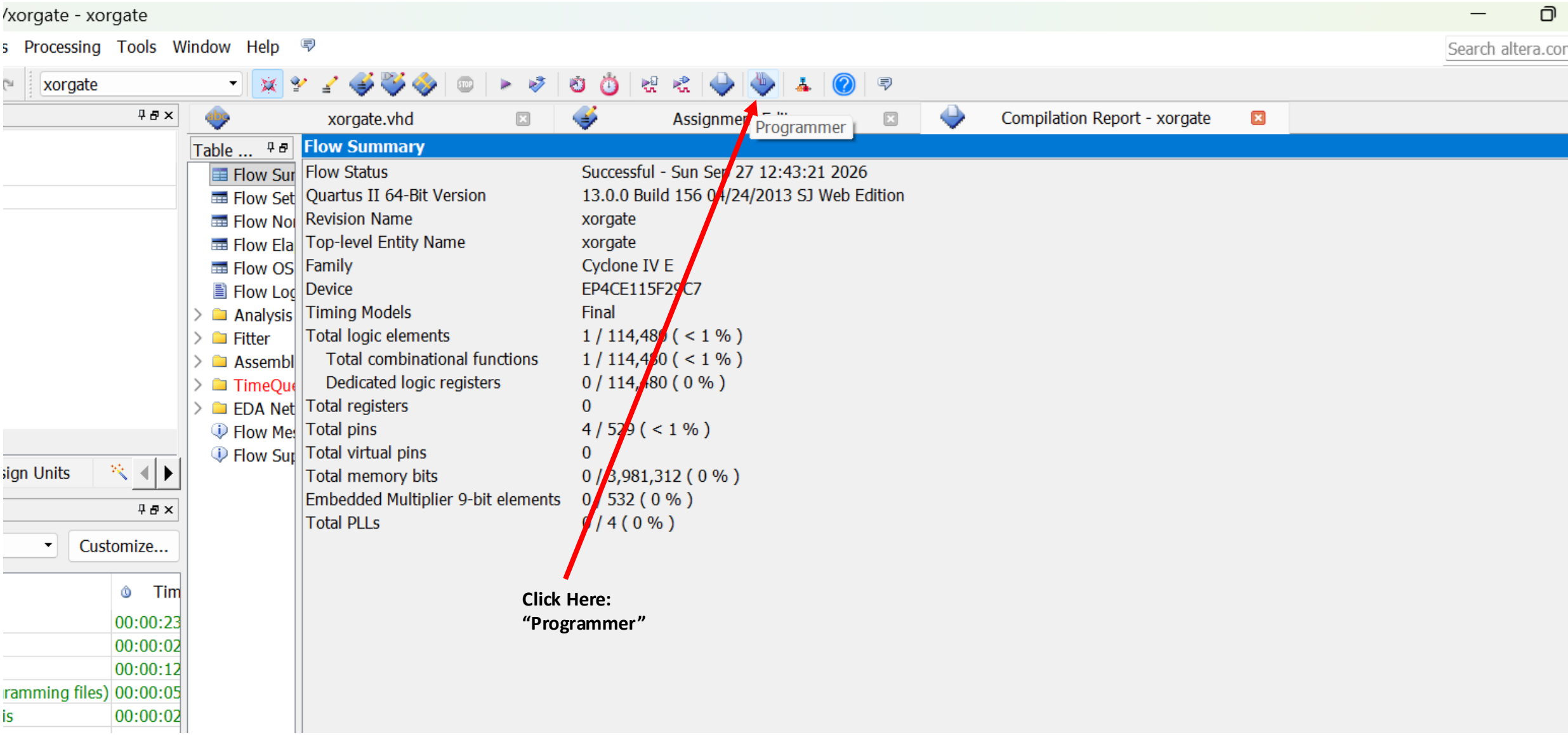
Flow Status	In progress - Sun Sep 27 12:43:02 2026
Quartus II 64-Bit Version	13.0.0 Build 156 04/24/2013 SJ Web Edition
Revision Name	xorgate
Top-level Entity Name	xorgate
Family	Cyclone IV E
Device	EP4CE115F29C7
Timing Models	Final
Total logic elements	1
Total combinational functions	1
Dedicated logic registers	0
Total registers	0
Total pins	4
Total virtual pins	0
Total memory bits	0
Embedded Multiplier 9-bit elements	0
Total PLLs	0

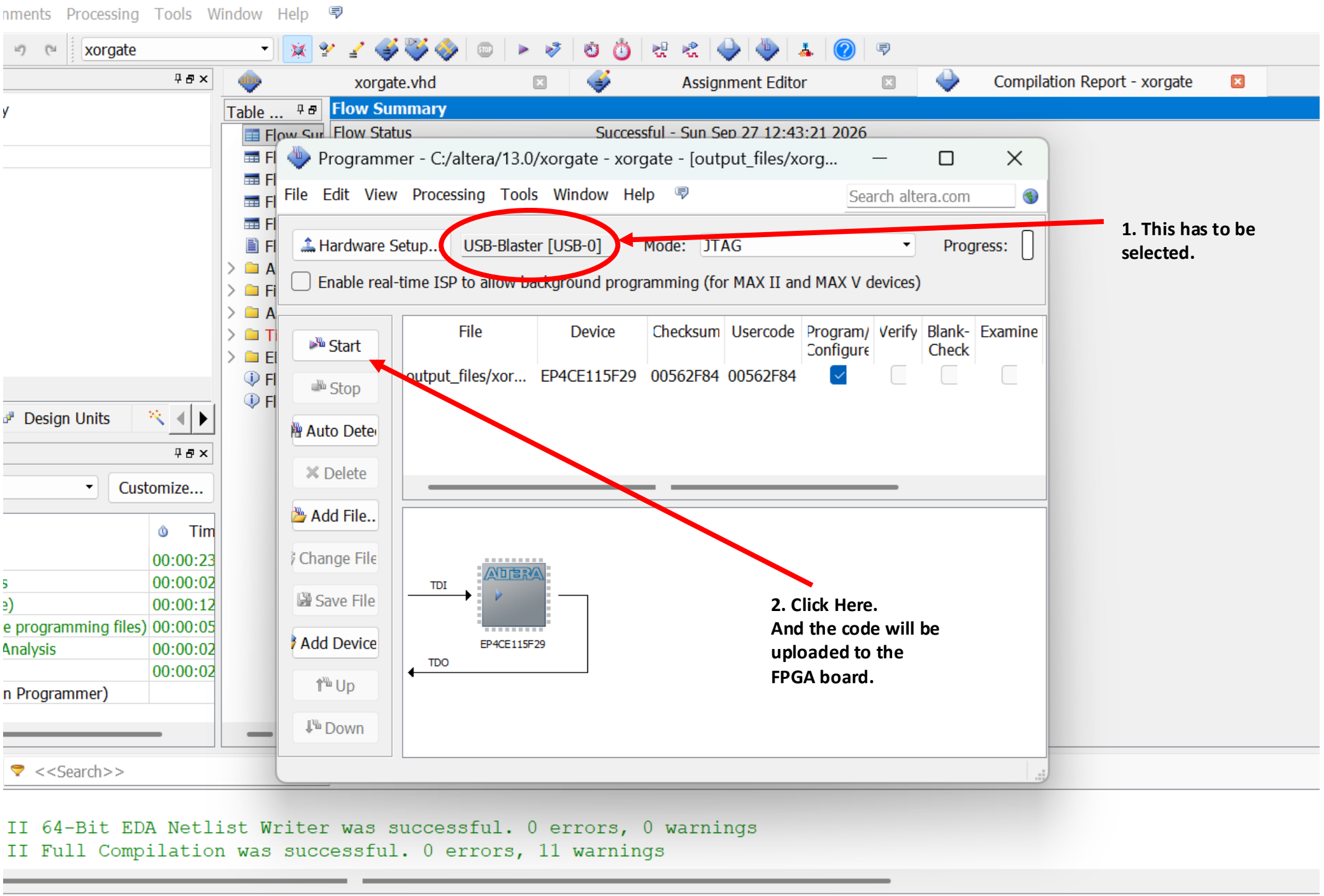
Compiling

All <<Search>>

Type	ID	Message
Info	176235	Finished register packing
Info	171121	Fitter preparation operations ending: elapsed time is 00:00:01


tus II 64-Bit - C:/altera/13.0/xorgate - xorgate
View Project Assignments Processing Tools Window Help
xorgate
vigator
Entity
e IV E: EP4CE115F29C7
gate
rchy
Files
Design Units
mpilation
Customize...
Task
Tim
Compile Design 00:00:23
Analysis & Synthesis 00:00:02
Fitter (Place & Route) 00:00:12
Assembler (Generate programming files) 00:00:05
TimeQuest Timing Analysis 00:00:02
EDA Netlist Writer 00:00:02
Program Device (Open Programmer)
xorgate.vhd
Assignment Editor
Compilation Report - xorgate
Table ...
Flow Sur
Flow Set
Flow Nor
Flow Ela
Flow OS
Flow Log
Analysis
Fitter
Assembl
TimeQue
EDA Net
Flow Me
Flow Sup
Flow Summary
Flow Status Successful - Sun Sep 27 12:43:21 2026
Quartus II 64-Bit Version 13.0.0 Build 156 04/24/2013 SJ Web Edition
Revision Name xorgate
Top-level Entity Name xorgate
Family Cyclone IV E
Device EP4CE115F29C7
Timing Models Final
Total logic elements 1 / 114,480 (< 1 %)
Total combinational functions 1 / 114,480 (< 1 %)
Dedicated logic registers 0 / 114,480 (0 %)
Total registers
Total pins
Total virtual pins
Total memory bits
Embedded Multiplier 9
Total PLLs
Quartus II
Full Compilation was successful (11 warnings)
OK
<<Search>>
ID
Message
Quartus II 64-Bit EDA Netlist Writer was successful. 0 errors, 0 warnings
293000 Quartus II Full Compilation was successful. 0 errors, 11 warnings


/xorgate - xorgate
Processing Tools Window Help
Search altera.com
xorgate
xorgate.vhd
Assignmer
Programmer
Compilation Report - xorgate
Table ...
Flow Sur
Flow Set
Flow Nor
Flow Ela
Flow OS
Flow Log
Analysis
Fitter
Assembl
TimeQue
EDA Net
Flow Me
Flow Sup
Flow Summary
Flow Status Successful - Sun Sep 27 12:43:21 2026
Quartus II 64-Bit Version 13.0.0 Build 156 04/24/2013 SJ Web Edition
Revision Name xorgate
Top-level Entity Name xorgate
Family Cyclone IV E
Device EP4CE115F29C7
Timing Models Final
Total logic elements 1 / 114,480 (< 1 %)
Total combinational functions 1 / 114,480 (< 1 %)
Dedicated logic registers 0 / 114,480 (0 %)
Total registers 0
Total pins 4 / 529 (< 1 %)
Total virtual pins 0
Total memory bits 0 / 3,981,312 (0 %)
Embedded Multiplier 9-bit elements 0 / 532 (0 %)
Total PLLs 0 / 4 (0 %)
ign Units
Customize...
Tim
00:00:23
00:00:02
00:00:12
ramming files) 00:00:05
is 00:00:02
Click Here:
"Programmer"


Programmer - C:/altera/13.0/xorgate - xorgate - [output_files/xorg...
File Edit View Processing Tools Window Help
Search altera.com
Hardware Setup..
USB-Blaster [USB-0]
Mode: JTAG
Progress:
Enable real-time ISP to allow background programming (for MAX II and MAX V devices)
Start
Stop
Auto Dete
Delete
Add File..
Change File
Save File
Add Device
Up
Down
File
Device
Checksum
Usercode
Program/ Configure
Verify
Blank-Check
Examine
output_files/xor...
EP4CE115F29
00562F84
00562F84
TDI
TDO
EP4CE115F29
1. This has to be selected.
2. Click Here.
And the code will be uploaded to the FPGA board.
xorgate.vhd
Assignment Editor
Compilation Report - xorgate
Flow Summary
Flow Status
Successful - Sun Sep 27 12:43:21 2026
Design Units
Customize...
<<Search>>
II 64-Bit EDA Netlist Writer was successful. 0 errors, 0 warnings
II Full Compilation was successful. 0 errors, 11 warnings

Quartus II 64-Bit - C:/altera/13.0/AND_GATE - AND_GATE

File Edit View Project Assignments Processing Tools Window Help

Search altera.com

Device...
Settings... Ctrl+Shift+E
TimeQuest Timing Analyzer Wizard...
Assignment Editor Ctrl+Shift+A
Pin Planner Ctrl+Shift+N
Remove Assignments...
Back-Annotate Assignments...
Import Assignments...
Export Assignments...
Assignment Groups...
LogicLock Regions Window Alt+L
Design Partitions Window Alt+D

Project Navigator

Cyclone IV E: EP4CE115F2
AND_GATE

Hierarchy Files Design Units IP Components

Tasks

Flow: Compilation Customize...

Task	Time
Compile Design	
Analysis & Synthesis	00:00:02
Fitter (Place & Route)	00:00:09
Assembler (Generate programming files)	00:00:03
TimeQuest Timing Analysis	00:00:01
EDA Netlist Writer	00:00:01
Program Device (Open Programmer)	

AND_GATE.vhd

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;

entity AND_GATE is
    Port (
        A : in  STD_LOGIC;
        B : in  STD_LOGIC;
        Y : out STD_LOGIC
    );
end AND_GATE;

architecture Behavioral of AND_GATE is
begin
    Y <= A AND B;
end Behavioral;
```

Compilation Report - AND_GATE

All <<Search>>

Type	ID	Message
	293000	Quartus II Full Compilation was successful. 0 errors, 12 warnings

Messages

System (1) Processing (125)

Edits pin assignments

100% 00:00:16

Activate Windows
Go to Settings to activate Windows.

Type here to search

31°C Rain showers

2:20 PM
10/30/2025

Pin Planner - C:/altera/13.0/AND_GATE - AND_GATE

File Edit View Processing Tools Window Help

Search altera.com

Report

Report not available

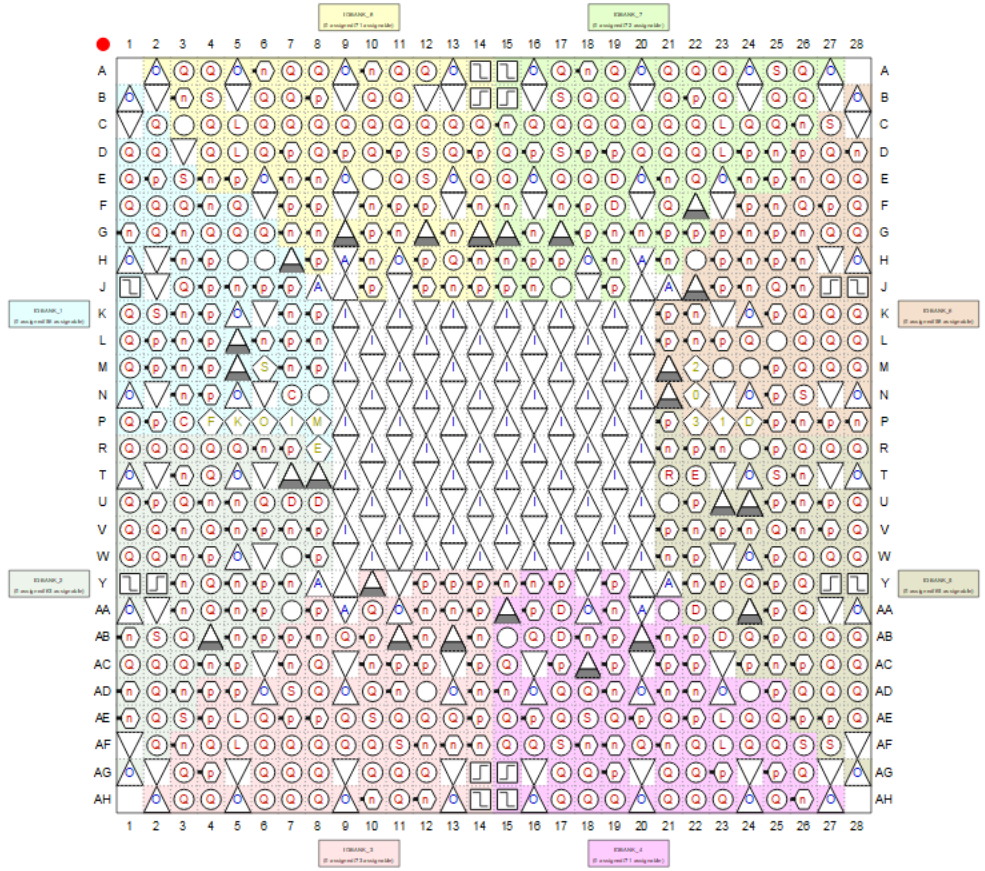
Groups Report

Tasks

Run Anal
Early Pin
Early
Run
Expo
Change
Show
Show
Show
Show
Show

Top View - Wire Bond

Cyclone IV E - EP4CE115F29C7



Named: * Edit: Filter: Pins: all

Node Name	Direction	Location	I/O Bank	VREF Group	Fitter Location	I/O Standard	Reserved	Current Strength	Slew Rate	Differential Pair
A	Input				PIN_AB5	2.5 V (default)		8mA (default)		
B	Input				PIN_AB6	2.5 V (default)		8mA (default)		
Y	Output				PIN_AC4	2.5 V (default)		8mA (default)	2 (default)	
<<new node>>										

All Pins

Activate Windows
Go to Settings to activate Windows.

0% 00:00:00

Type here to search

Rainy days ahead

2:21 PM
10/30/2025

Pin Planner - C:/altera/13.0/AND_GATE - AND_GATE

File Edit View Processing Tools Window Help

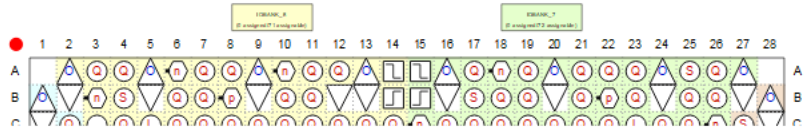
Search altera.com

Report

Report not available

Top View - Wire Bond

Cyclone IV E - EP4CE115F29C7



Quartus II 64-Bit - C:/altera/13.0/AND_GATE - AND_GATE

File Edit View Project Assignments Processing Tools Window Help

Search altera.com

AND_GATE

Project Navigator

Entity

Cyclone IV E: EP4CE115F29C7

AND_GATE

Hierarchy Files Design Units IP Components

Tasks

Flow: Compilation Customize...

Task	Time
Compile Design	
Analysis & Synthesis	00:00:02
Fitter (Place & Route)	00:00:09
Assembler (Generate programming files)	00:00:03
TimeQuest Timing Analysis	00:00:01
EDA Netlist Writer	00:00:01
Program Device (Open Programmer)	

AND_GATE.vhd

Compilation Report - AND_GATE

```
1  library IEEE;
2  use IEEE.STD_LOGIC_1164.ALL;
3
4  entity AND_GATE is
5      Port (
6          A : in  STD_LOGIC;
7          B : in  STD_LOGIC;
8          Y : out STD_LOGIC
9      );
10 end AND_GATE;
11
12 architecture Behavioral of AND_GATE is
13 begin
14     Y <= A AND B;
15 end Behavioral;
```

Pin Planner - C:/altera/13.0/AND_GATE - AND_GATE

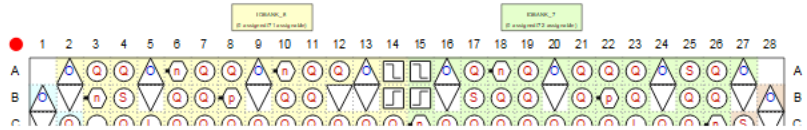
File Edit View Processing Tools Window Help

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Report

Report not available

Top View - Wire Bond
Cyclone IV E - EP4CE115F29C7



Quartus II 64-Bit - C:/altera/13.0/AND_GATE - AND_GATE

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AND_GATE

Project Navigator

Entity

Cyclone IV E: EP4CE115F29C7

AND_GATE

Hierarchy Files Design Units IP Components

Tasks

Flow: Compilation Customize...

	Task	Time
?	Compile Design	
?	Analysis & Synthesis	00:00:02
?	Fitter (Place & Route)	00:00:09
?	Assembler (Generate programming files)	00:00:03
?	TimeQuest Timing Analysis	00:00:01
?	EDA Netlist Writer	00:00:01
	Program Device (Open Programmer)	

AND_GATE.vhd

Compilation Report - AND_GATE

```
1   library IEEE;
2   use IEEE.STD_LOGIC_1164.ALL;
3
4   entity AND_
5       Port (
6           A :
7           B :
8           Y :
9       );
10  end AND_GAT
11
12  architectur
13  begin
14      Y <= A
15  end Behavio
```

Programmer - C:/altera/13.0/AND_GATE - AND_GATE - [output_files/AND_GATE.cdf]

File Edit View Processing Tools Window Help

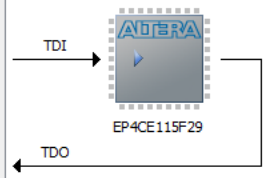
Search altera.com

Hardware Setup... No Hardware Mode: JTAG Progress:

Enable real-time ISP to allow background programming (for MAX II and MAX V devices)

Start
Stop
Auto Detect
Delete
Add File...
Change File...
Save File
Add Device...
Up
Down

File	Device	Checksum	Usercode	Program/ Configure	Verify	Blank-Check	Examine
output_files/AND_GATE.sof	EP4CE115F29	00562D08	00562D08	☑	☐	☐	☐



Pin Planner - C:/altera/13.0/AND_GATE - AND_GATE

File Edit View Processing Tools Window Help

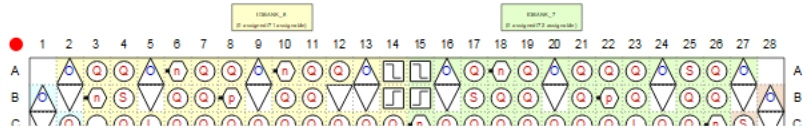
Search altera.com

Report

Report not available

Top View - Wire Bond

Cyclone IV E - EP4CE115F29C7



Quartus II 64-Bit - C:/altera/13.0/AND_GATE - AND_GATE

File Edit View Project Assignments Processing Tools Window Help

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AND_GATE

Project Navigator

Entity

Cyclone IV E: EP4CE115F29C7

AND_GATE

Hierarchy Files Design Units IP Components

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	EDA Netlist Writer	00:00:01
	Program Device (Open Programmer)	

AND_GATE.vhd

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Programmer - C:/altera/13.0/AND_GATE - AND_GATE - [output_files/AND_GATE.cdf]

File Edit View Processing Tools Window Help

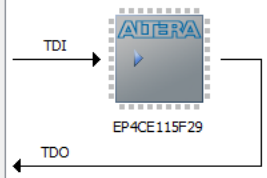
Search altera.com

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Enable real-time ISP to allow background programming (for MAX II and MAX V devices)

Start Stop Auto Detect Delete Add File... Change File... Save File Add Device... Up Down

File	Device	Checksum	Usercode	Program/ Configure	Verify	Blank-Check	Examine
output_files/AND_GATE.sof	EP4CE115F29	00562D08	00562D08	☑	☐	☐	☐



Hardware Setup

Hardware Settings JTAG Settings

Select a programming hardware setup to use when programming devices. This programming hardware setup applies only to the current programmer window.

Currently selected hardware: No Hardware

Available hardware items

Hardware	Server	Port

Add Hardware... Remove Hardware

Close

Pin Planner - C:/altera/13.0/AND_GATE - AND_GATE

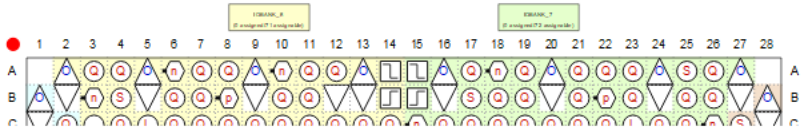
File Edit View Processing Tools Window Help

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Report

Report not available

Top View - Wire Bond
Cyclone IV E - EP4CE115F29C7



Quartus II 64-Bit - C:/altera/13.0/AND_GATE - AND_GATE

File Edit View Project Assignments Processing Tools Window Help

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AND_GATE

Project Navigator

Entity

Cyclone IV E: EP4CE115F29C7

AND_GATE

Hierarchy Files Design Units IP Components

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AND_GATE.vhd

```
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6          A :
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13 begin
14     Y <= A
15 end Behavio
```

Compilation Report - AND_GATE

Programmer - C:/altera/13.0/AND_GATE - AND_GATE - [output_files/AND_GATE.cdf]

File Edit View Processing Tools Window Help

Search altera.com

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File	Device	Checksum	Usercode	Program/ Configure	Verify	Blank-Check	Examine
output_files/AND_GATE.sof	EP4CE115F29	00562D08	00562D08	☑	☐	☐	☐

